MILITARY SPECIFICATION

* MODULES, STANDARD ELECTRONIC

INVERTER, DIGITAL, KEY CODE RBG

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic digital inverter module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table II.

Name	Inverter
Family	Digital
Key code	RBG
Size	1A
Weight	35 grams maximum
Failure rate	0.788 failures/10^6 hours maximum
* Environment	Class I as specified in MIL-STD-1389
Supply voltage	5 Vdc ±10 percent
Maximum power dissipation	
All outputs at low level	891 mW (outputs not loaded)
All outputs at high level	396 mW (outputs not loaded)
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current	
Low level (V_{IL} = 0.0 V)	-2.2 mA maximum
(V_{IL} = 0.5 V)	-2.0 mA maximum
High level (V_{IH} = 2.5 V)	50.0 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN ELEX 5043, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Output voltage	
Low level (I_{OL} = 20 mA)	0.5 V maximum
High level (I_{OH} = -1 mA)	2.5 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.5 V)	20 mA
High level (V_{OH} = 2.5 V)	-1.0 mA
* Propagation delay time (t_{PLH})	
(50 pF load)	2.0 ns minimum, 8.0 ns maximum
(150 pF load)	4.0 ns minimum, 12.0 ns maximum
Propagation delay time (t_{PHL})	
(50 pF load)	2.0 ns minimum, 8.0 ns maximum
(150 pF load)	4.0 ns minimum, 12.0 ns maximum
Delta propagation delay time (Δt_{PLH}) (50 pF load)	
Between inverters 1, 2, 3	3.0 ns maximum
Between inverters 4, 5, 6	3.0 ns maximum
Between inverters 7, 8, 9	3.0 ns maximum
Between inverters 10, 11, 12	3.0 ns maximum
Between inverters 13, 14, 15	3.0 ns maximum
Between inverters 16, 17, 18	3.0 ns maximum
Delta propagation delay time (Δt_{PHL}) (50 pF load)	
Between inverters 1, 2, 3	3.0 ns maximum
Between inverters 4, 5, 6	3.0 ns maximum
Between inverters 7, 8, 9	3.0 ns maximum
Between inverters 10, 11, 12	3.0 ns maximum
Between inverters 13, 14, 15	3.0 ns maximum
Between inverters 16, 17, 18	3.0 ns maximum

* 1.3 Module function. This module contains 18 independent logic inverters.

1.3.1 Input to output relationship. Each inverter has the input to output relationship of table I for positive logic.

TABLE I. Input to output relationship.

Input (A)	Output (B)
L	H
H	L

NOTES:
1. H = High-voltage level.
2. L = Low-voltage level.

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1.3.2 Parallel outputs. Paralleling of outputs to obtain extended drive is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The circuit groups which can be paralled are:

a. Circuits 1, 2, and 3.
b. Circuits 4, 5, and 6.
c. Circuits 7, 8, and 9.
d. Circuits 10, 11, and 12.
e. Circuits 13, 14, and 15.
f. Circuits 16, 17, and 18.

1.3.3 Ground pins. All electrical parameters are measured with pins 3, 10, and 18 connected separately to 0 volt return. System problems may result from a different grounding method.

1.3.4 Crosstalk relationships. The module has the following crosstalk relationships.

1.3.4.1 Module crosstalk. When connected in the worst case mode of operation (all outputs but one switching simultaneously from a high level to a low level), the RBG module could exhibit substantial crosstalk. When the quiet line is a low level, there may be as much as an 800 mV positive going spike coincident with the negative edge of the switching outputs.

1.3.4.2 Circuit crosstalk. The crosstalk on any output pin is nearly directly proportional to the number of switching outputs. For every switching output in group A (circuits 1-3) the noise induced on a low level quiet output in group A is as much as 137 mV. For every switching output outside group A (circuits 4-18), the noise induced on a low level quiet output in group A is as much as 35 mV. The preceeding relationships also hold for group B (circuits 4-6), group C (circuits 7-9), group D (circuits 10-12), group E (circuits 13-15) and group F (circuits 16-18).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

* MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

* MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
* MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.788 failures per million operating hours when predicted in accordance with MIL-STD-1389.

* 3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be RBG.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

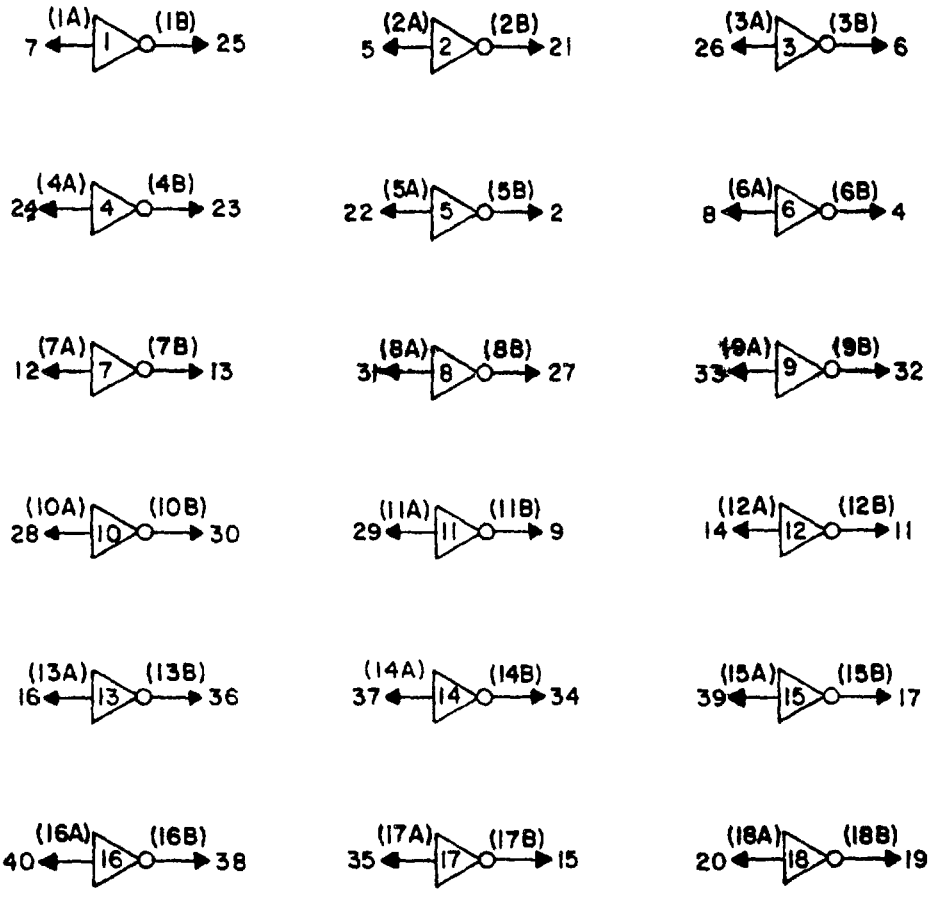
3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/82-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II (see 6.2).

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NOTES:
1. 1A-18A = Inverter inputs.
2. 1B-18B = Inverter outputs.

FIGURE 1. RBG module function diagram, digital inverter

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
B1-B18	V_{OUT} (low-loaded)	IV	S	0.01	1.0	0.01	1.0	0.01	1.0	0.01	1.0	V
C1-C18	Input breakdown current	V	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
D1-D18	Output leakage current	VI	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
E1-E18	Output short circuit current	VII	X	-40	-100	-40	-100	-40	-100	-40	-100	mA
F1-F18	Input clamp diode voltage	VIII	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
G1-G18	Static output voltage, low level	IX	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
H1-H18	Static output voltage, high level	X	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
J1-J18	Input current, high level	XI	X	0	50	0	50	0	50	0	50	µA
K1-K18	Input current, low level	XII	X	-0.4	-2.2	-0.4	-2.2	-0.4	-2.2	-0.4	-2.2	mA
L1, L2	Ground pin continuity	XIII	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ω
L3, L4	Ground pin continuity	XIII	Q	0.0	0.10	0.0	0.10	0.0	0.10	0.0	0.10	Ω
M1	Power supply current, low level	XIV	S	30	162	30	162	30	162	30	162	mA
M2	Power supply current, high level	XIV	S	15	72	15	72	15	72	15	72	mA
N1-N18	Transition time (t_{TLH}), C_L = 50 pF	XV	S	2.0	15.0	2.0	15.0	2.0	18.0	2.0	18.0	ns
N19-N36	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	2.0	27.0	2.0	27.0	2.0	30.0	2.0	30.0	ns
P1-P18	Transition time (t_{THL}), C_L = 50 pF	XV	S	2.0	6.0	2.0	6.0	2.0	8.0	2.0	8.0	ns
P19-P36	Transition time (t_{THL}), C_L = 150 pF	XV	Q	2.0	11.0	2.0	11.0	2.0	13.0	2.0	13.0	ns
* Q1-Q18	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	2.0	7.0	2.0	7.0	2.0	8.0	2.0	8.0	ns

See footnotes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q19-Q36	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	4.0	10.5	4.0	10.5	4.0	12.0	4.0	12.0	ns
R1-R18	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	2.0	7.5	2.0	7.5	2.0	8.0	2.0	8.0	ns
R19-R36	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	4.0	11.5	4.0	11.5	4.0	12.0	4.0	12.0	ns
S1-S18	Output to output delta propagation delay time (Δt_{PLH}), C_L = 50 pF	XVI	S	-3.0	3.0	-3.0	3.0	-3.0	3.0	-3.0	3.0	ns
T1-T18	Output to output delta propagation delay time (Δt_{PHL}), C_L = 50 pF	XVI	S	-3.0	3.0	-3.0	3.0	-3.0	3.0	-3.0	3.0	ns
U1-U18	Output crosstalk, low level	XVII	Q	0	800	0	800	0	800	0	800	mV
U19-U36	Output crosstalk, low level	XVII	Q	0	525	0	525	0	525	0	525	mV
V1-V18	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
W	Power supply transient amplitude	XVIII	Q	0	1.0	0	1.0	0	1.0	0	1.0	V
X	Power supply filter capacitance	XIX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	uF
Y1-Y18	Input capacitance	XX	Q	5.0	15.0	5.0	15.0	5.0	15.0	5.0	15.0	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
DC ammeter	Item X01	1
DC voltmeter	Item V03	1
Pulse generator	Item J05	1
Pulse generator	Item J01	1
DC power supply	Item P01	4
Megohm bridge	Item R06	1
Resistance bridge	Item R02	1
Capacitance bridge	Item C01	1
Capacitance meter	Item C05	1
Resistors		
249Ω ±1%, 1/4 W		18
590Ω ±1%, 1/4 W		1
51.1Ω ±1%, 1/4 W		1
51Ω ±5%, 1/2 W		2
2.49 kΩ ±1%, 1/8 W		18
1.0 kΩ ±1%, 1/8 W		1
Value as required for figure 12, note 2, 1/4 W		18
Capacitors		
Value as required for figure 12, note 4, test type S		18
Value as required for figure 12, note 4, test type Q		1
Diode		
1N3600, 1N4150, or equivalent		18

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II.

* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

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4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. If conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table II which are marked with an "X" in the test type column, and test condition X, shall be performed using the initial limits at 25° ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through X shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table II. Following the return of the module to 25°C, electrical tests D through Y shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table II.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/82-1 supersedes part number M28787/82. The two parts are interchangeable.

6.4 0 volt reference. The reference for test setups will be 0 volt and not necessarily earth ground.

6.5 Decoupling. Decoupling of test voltages and V_{CC} is permissible except for test condition W, power supply transient amplitude test.

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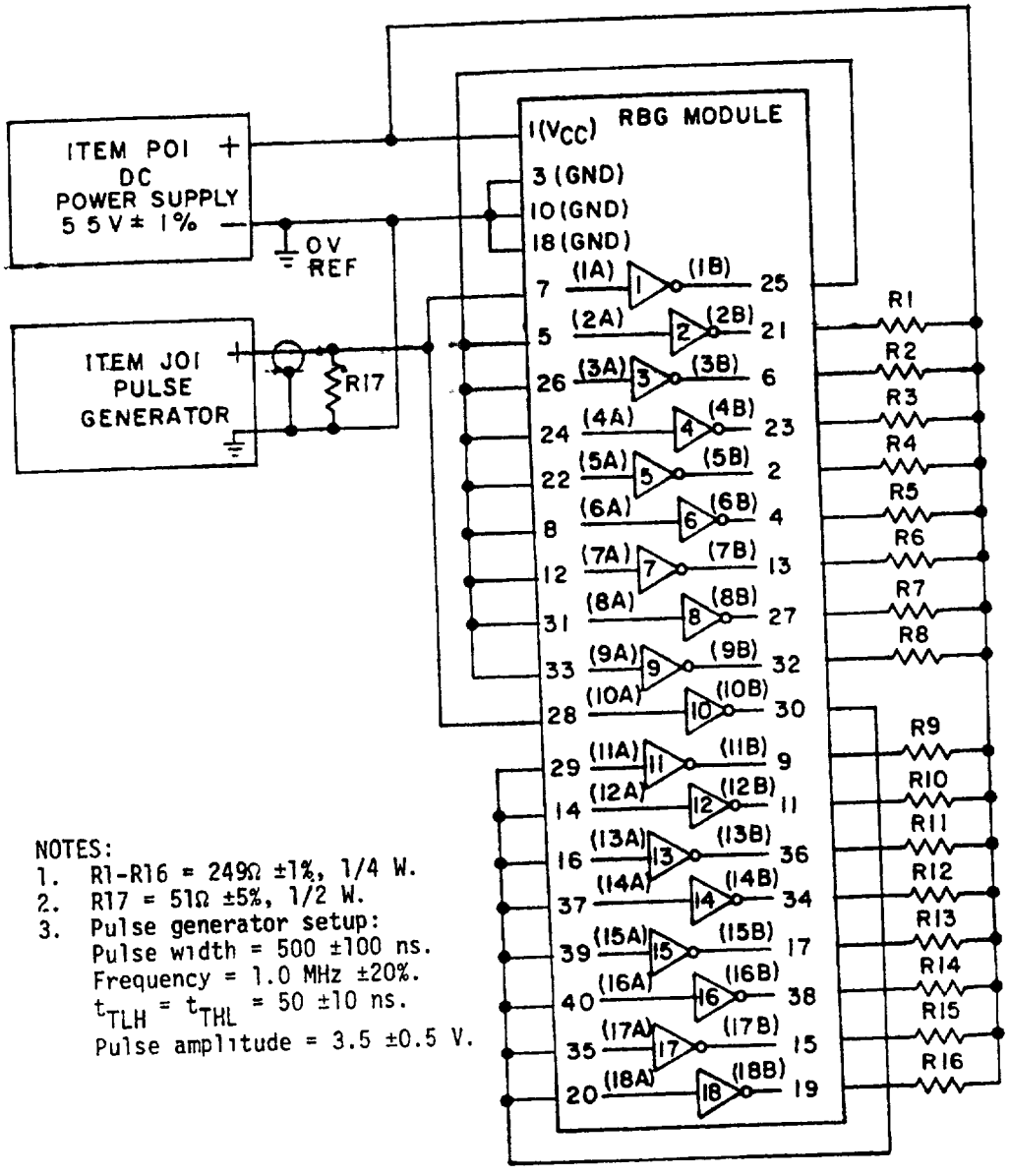
* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, correction, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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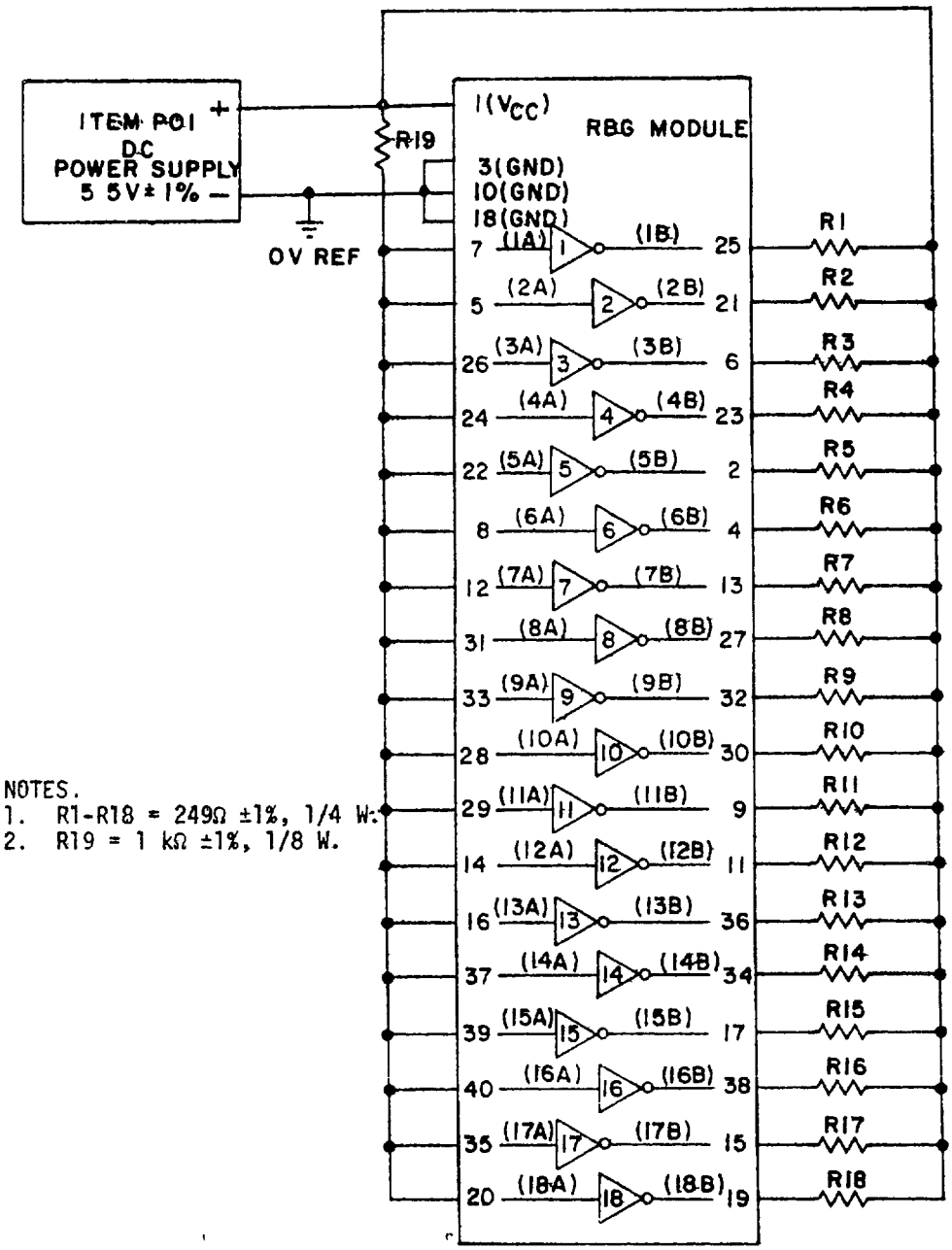
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NOTES:
1. R1-R16 = 249Ω ±1%, 1/4 W.
2. R17 = 51Ω ±5%, 1/2 W.
3. Pulse generator setup:
   Pulse width = 500 ±100 ns.
   Frequency = 1.0 MHz ±20%.
   $t_{TLH} = t_{THL} = 50 \pm 10$ ns.
   Pulse amplitude = 3.5 ±0.5 V.

FIGURE 2. Operating temperature and life test circuit.

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NOTES.
1. R1-R18 = 249Ω ±1%, 1/4 W.
2. R19 = 1 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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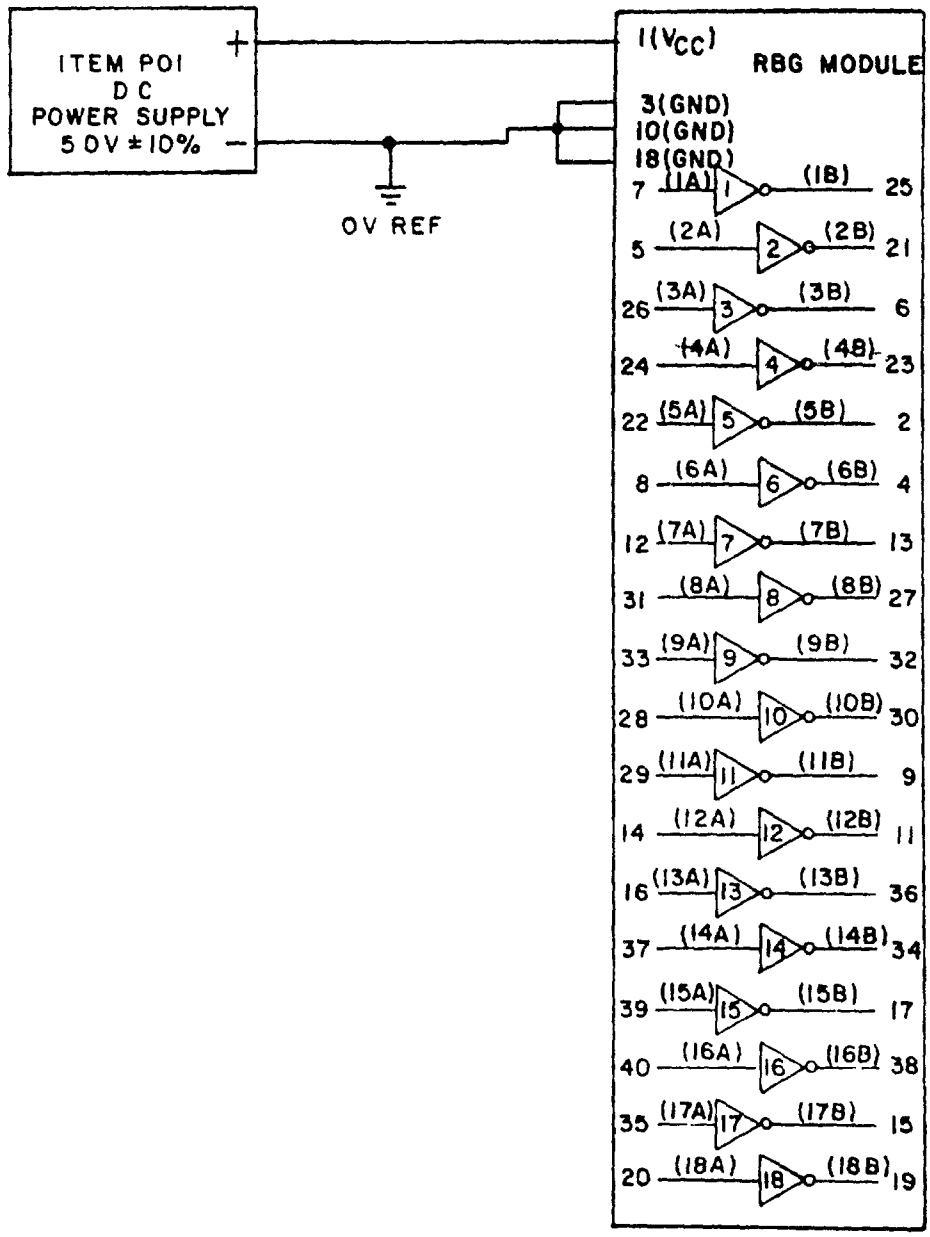


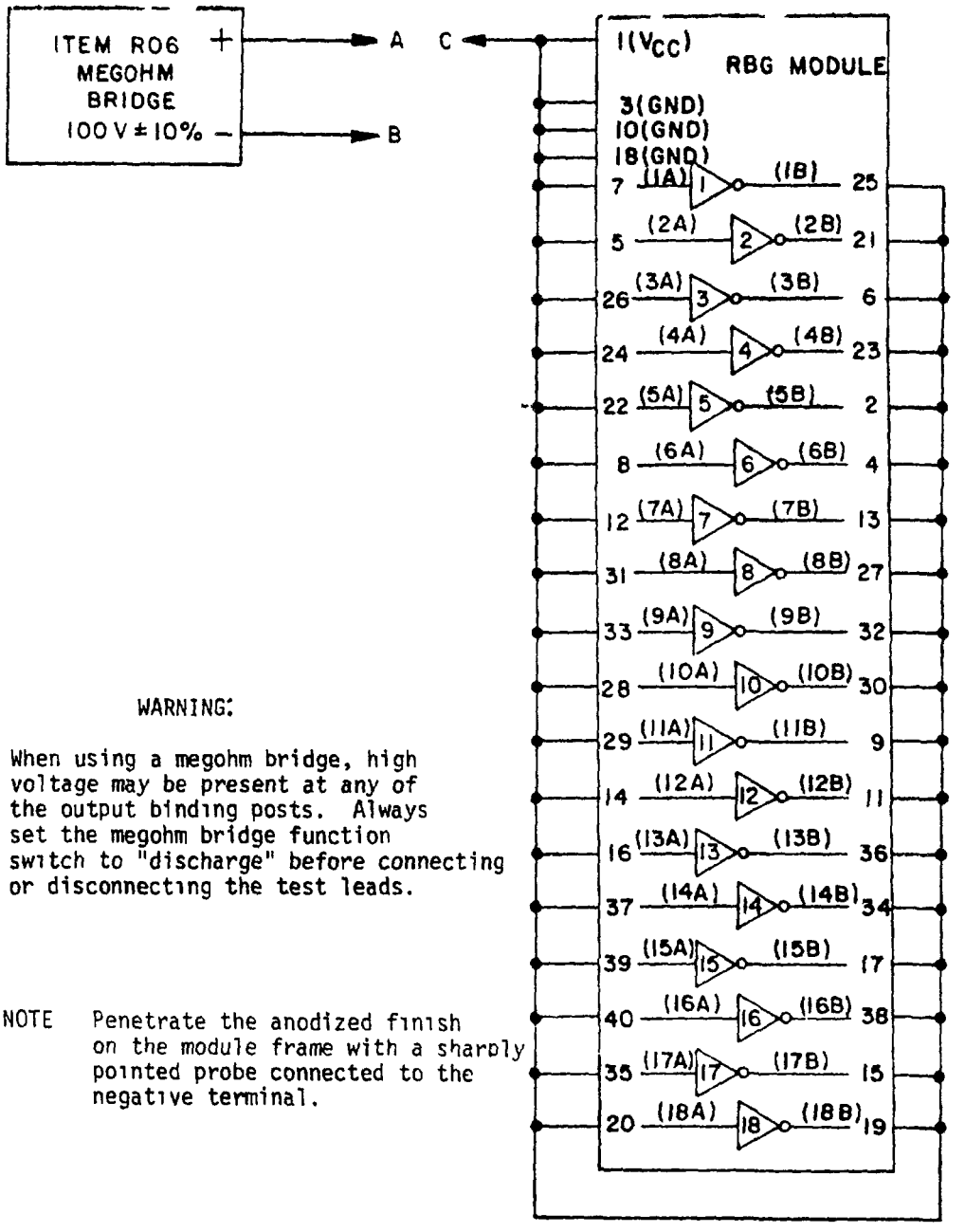
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	5

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

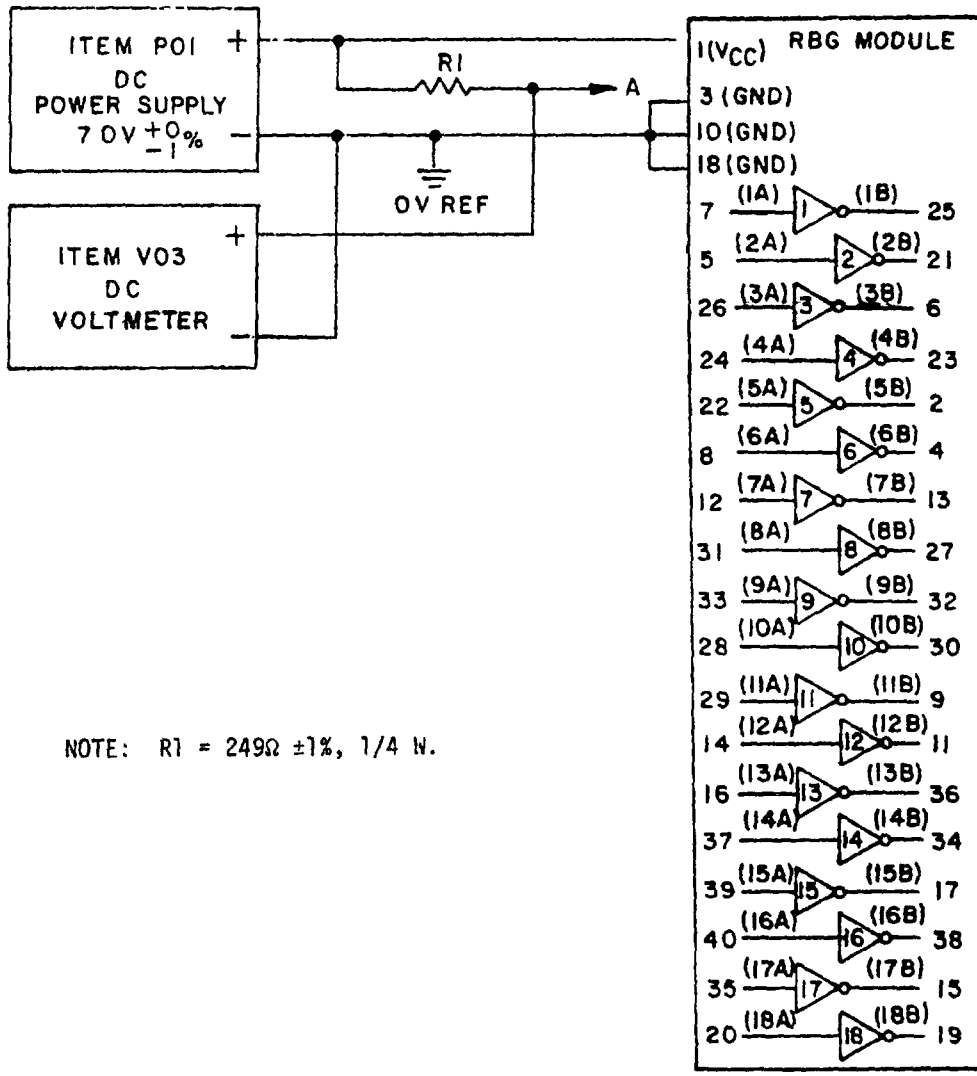
FIGURE 5. Isolation test circuit for table III.

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TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
B1	1B V_{OUT}(low-loaded)	25	0.01	1.0	V	6
B2	2B	21				
B3	3B	6				
B4	4B	23				
B5	5B	2				
B6	6B	4				
B7	7B	13				
B8	8B	27				
B9	9B	32				
B10	10B	30				
B11	11B	9				
B12	12B	11				
B13	13B	36				
B14	14B	34				
B15	15B	17				
B16	16B	38				
B17	17B	15				
B18	18B V_{OUT}(low-loaded)	19	0.01	1.0	V	6

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NOTE: R1 = 249Ω ±1%, 1/4 W.

FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	1A	Input breakdown current	5,8,12,14,16,20,22,24,26, 28,29,31,33,35,37,39,40	7	0.0	1.0	mA	7
C2	2A		7,8,12,14,16,20,22,24,26, 28,29,31,33,35,37,39,40	5				
C3	3A		5,7,8,12,14,16,20,22,24, 28,29,31,33,35,37,39,40	26				
C4	4A		5,7,8,12,14,16,20,22,26, 28,29,31,33,35,37,39,40	24				
C5	5A		5,7,8,12,14,16,20,24,26,28, 29,31,33,35,37,39,40	22				
C6	6A		5,7,12,14,16,20,22,24,26, 28,29,31,33,35,37,39,40	8				
C7	7A		5,7,8,14,16,20,22,24,26,28, 29,31,33,35,37,39,40	12				
C8	8A		5,7,8,12,14,16,20,22,24,26, 28,29,33,35,37,39,40	31				
C9	9A		5,7,8,12,14,16,20,22,24,26, 28,29,31,35,37,39,40	33				
C10	10A		5,7,8,12,14,16,20,22,24,26, 29,31,33,35,37,39,40	28				
C11	11A		5,7,8,12,14,16,20,22,24,26, 28,31,33,35,37,39,40	29				
C12	12A	Input breakdown current	5,7,8,12,16,20,22,24,26,28, 29,31,33,35,37,39,40	14	0.0	1.0	mA	7

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° +5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C13	13A Input breakdown current	5,7,8,12,14,20,22,24,26,28, 29,31,33,35,37,39,40	16	0.0	1.0	mA	7
C14	14A	5,7,8,12,14,16,20,22,24,26, 28,29,31,33,35,39,40	37				
C15	15A	5,7,8,12,14,16,20,22,24,26, 28,29,31,33,35,37,40	39				
C16	16A	5,7,8,12,14,16,20,22,24,26, 28,29,31,33,35,37,39	40				
C17	17A	5,7,8,12,14,16,20,22,24,26, 28,29,31,33,37,39,40	35				
C18	18A Input breakdown current	5,7,8,12,14,16,22,24,26,28, 29,31,33,35,37,39,40	20	0.0	1.0	mA	7

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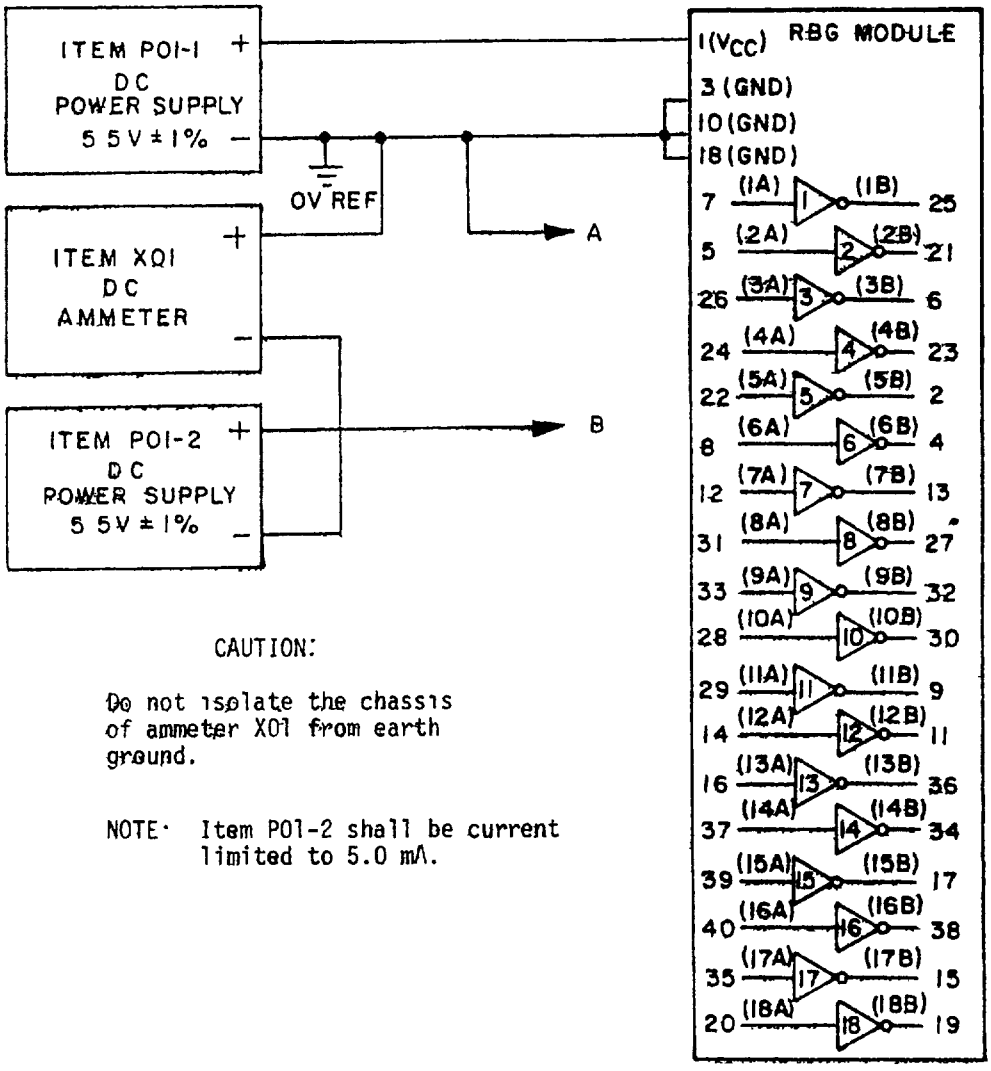


FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Output leakage current test.

Test condition	Parameter	Connect	Initial limits 25° +5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
D1	1B Output leakage current	25	0.0	1.0	mA	8
D2	2B	21				
D3	3B	6				
D4	4B	23				
D5	5B	2				
D6	6B	4				
D7	7B	13				
D8	8B	27				
D9	9B	32				
D10	10B	30				
D11	11B	9				
D12	12B	11				
D13	13B	36				
D14	14B	34				
D15	15B	17				
D16	16B	38				
D17	17B	15				
D18	18B Output leakage current	19	0.0	1.0	mA	8

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Check the source to verify that this is the current version before use.

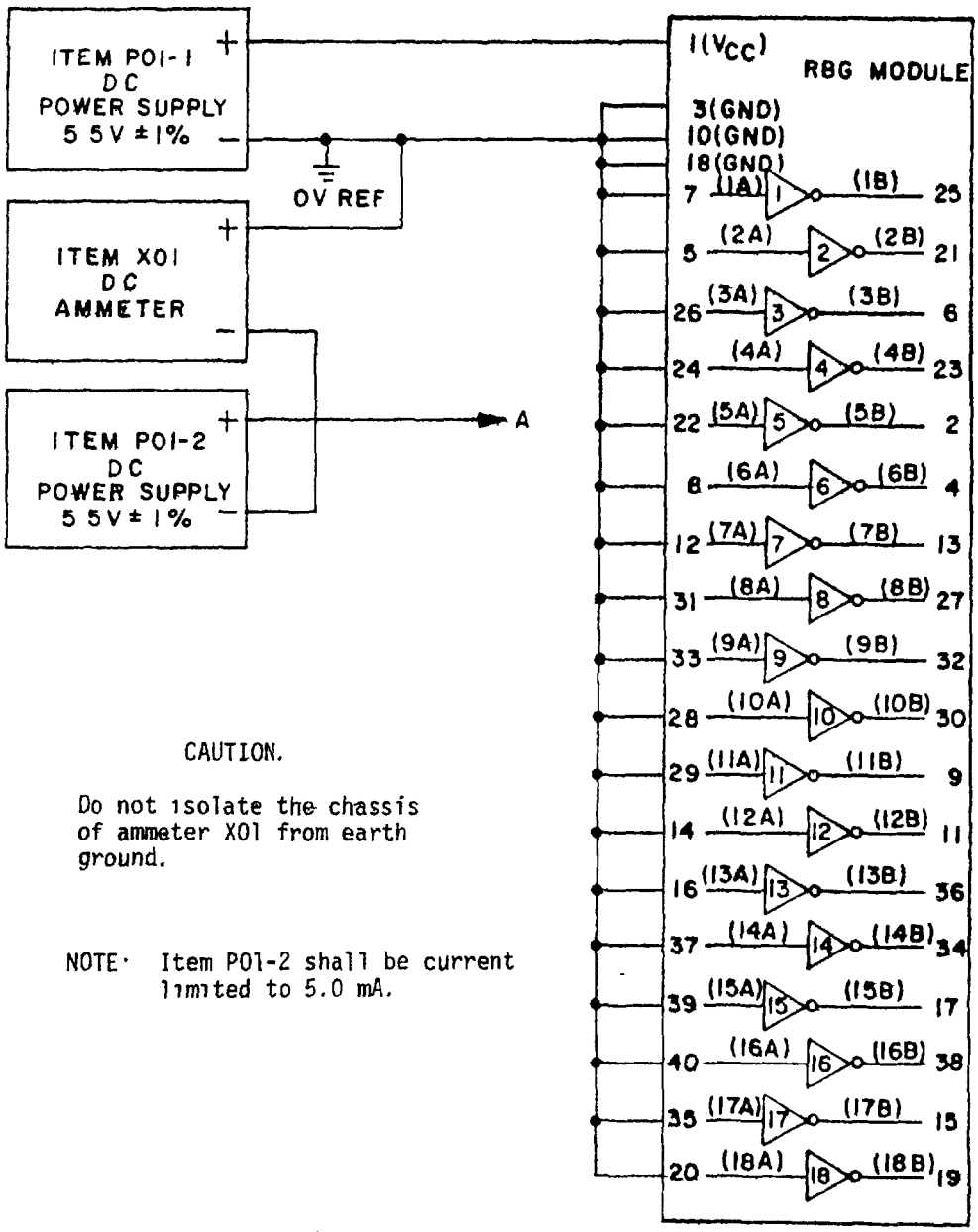


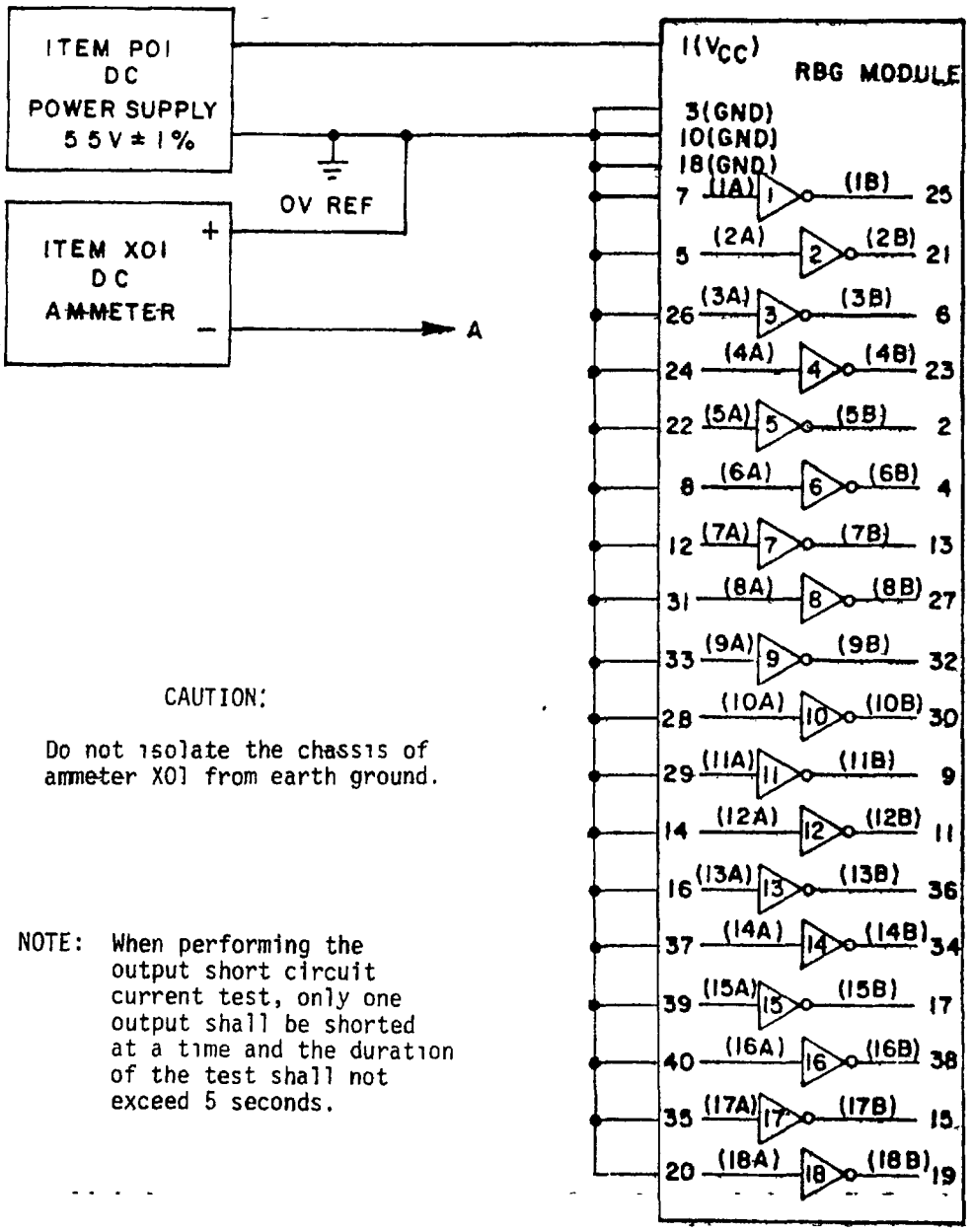
FIGURE 8. Output leakage current test circuit for table VI.

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TABLE VII. Output short circuit current test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
E1	1B	Output short circuit current	25	-40	-100	mA	9
E2	2B		21				
E3	3B		6				
E4	4B		23				
E5	5B		2				
E6	6B		4				
E7	7B		13				
E8	8B		27				
E9	9B		32				
E10	10B		30				
E11	11B		9				
E12	12B		11				
E13	13B		36				
E14	14B		34				
E15	15B		17				
E16	16B		38				
E17	17B		15				
E18	18B	Output short circuit current	19	-40	-100	mA	9

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: When performing the output short circuit current test, only one output shall be shorted at a time and the duration of the test shall not exceed 5 seconds.

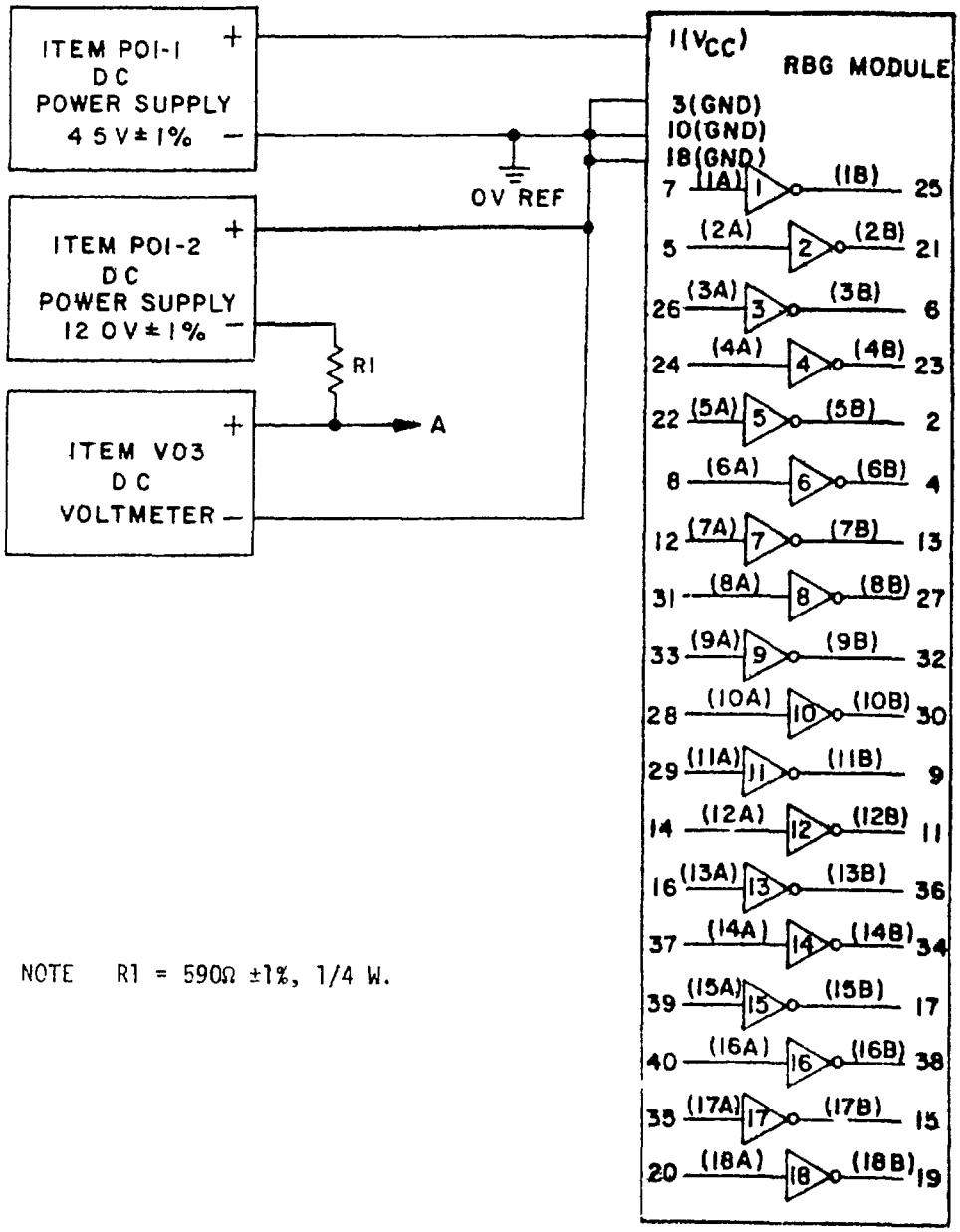
FIGURE 9. Output short circuit current test circuit for table VII.

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TABLE VIII. Input clamp diode voltage test.

Test condition		Parameter	Connect	Initial limits 25° +5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
F1	1A	Input clamp diode voltage	7	-0.4	-1.2	V	10
F2	2A		5				
F3	3A		26				
F4	4A		24				
F5	5A		22				
F6	6A		8				
F7	7A		12				
F8	8A		31				
F9	9A		33				
F10	10A		28				
F11	11A		29				
F12	12A		14				
F13	13A		16				
F14	14A		37				
F15	15A		39				
F16	16A		40				
F17	17A		35				
F18	18A	Input clamp diode voltage	20	-0.4	-1.2	V	10

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NOTE R1 = 590Ω ±1%, 1/4 W.

FIGURE 10. Input clamp diode voltage test circuit for table VIII.

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TABLE IX. Static output voltage (low-level) test.

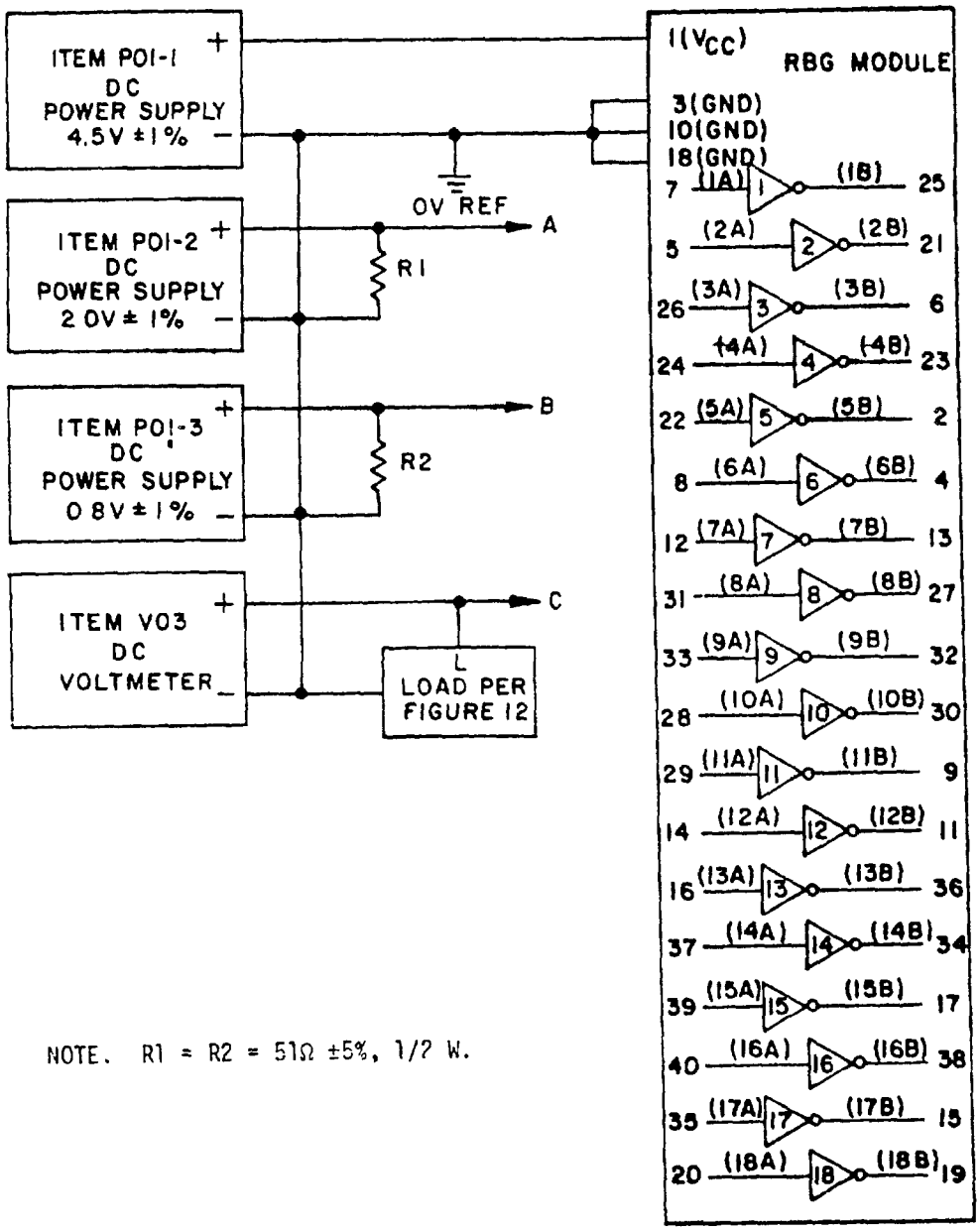
Test condition	Parameter		Connect			Initial limits 25° +5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	1B	Static output voltage, low level	7	Open	25	0.0	0.5	V	11
G2	2B		5		21				
G3	3B		26		6				
G4	4B		24		23				
G5	5B		22		2				
G6	6B		8		4				
G7	7B		12		13				
G8	8B		31		27				
G9	9B		33		32				
G10	10B		28		30				
G11	11B		29		9				
G12	12B		14		11				
G13	13B		16		36				
G14	14B		37		34				
G15	15B		39		17				
G16	16B		40		38				
G17	17B		35		15				
G18	18B	Static output voltage, low level	20	Open	19	0.0	0.5	V	11

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TABLE X. Static output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1B Static output voltage, high level	Open	7	25	2.5	4.5	V	11
H2	2B		5	21				
H3	3B		26	6				
H4	4B		24	23				
H5	5B		22	2				
H6	6B		8	4				
H7	7B		12	13				
H8	8B		31	27				
H9	9B		33	32				
H10	10B		28	30				
H11	11B		29	9				
H12	12B		14	11				
H13	13B		16	36				
H14	14B		37	34				
H15	15B		39	17				
H16	16B		40	38				
H17	17B		35	15				
H18	18B Static output voltage, high level	Open	20	19	2.5	4.5	V	11

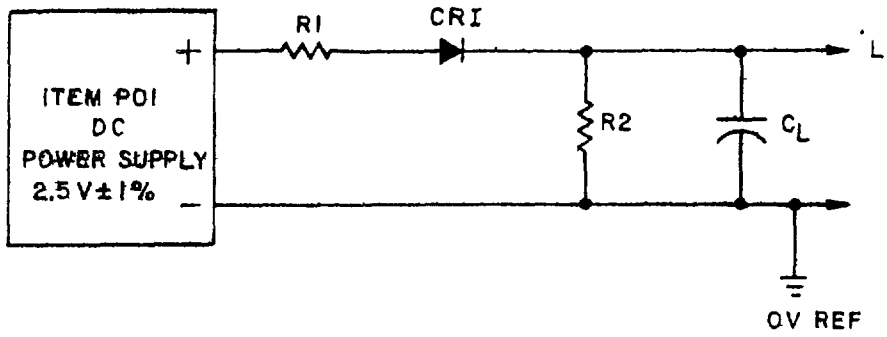
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NOTE. R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 11. Static output voltage (low-level and high-level) test circuit for tables IX and X.

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NOTES
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 20 mA ±5% with point L held at 0.5 V
3. R2 = 2.49 kΩ ±1%, 1/8 W.
4. Test type S:
   C_L = 50 ±5 pF (including probe and parasitic capacitance).
   Test type Q.
   C_L = 150 ±5 pF (including probe and parasitic capacitance).
5. Capacitor C_L is not required for dc electrical test conditions G1-G18 and H1-H18.
6. Lead length from load circuit to module output pin under test is to be less than 1.00 inch (25.4 mm)

FIGURE 12. RGB logic load circuit.

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TABLE XI. Input current (high-level) test.

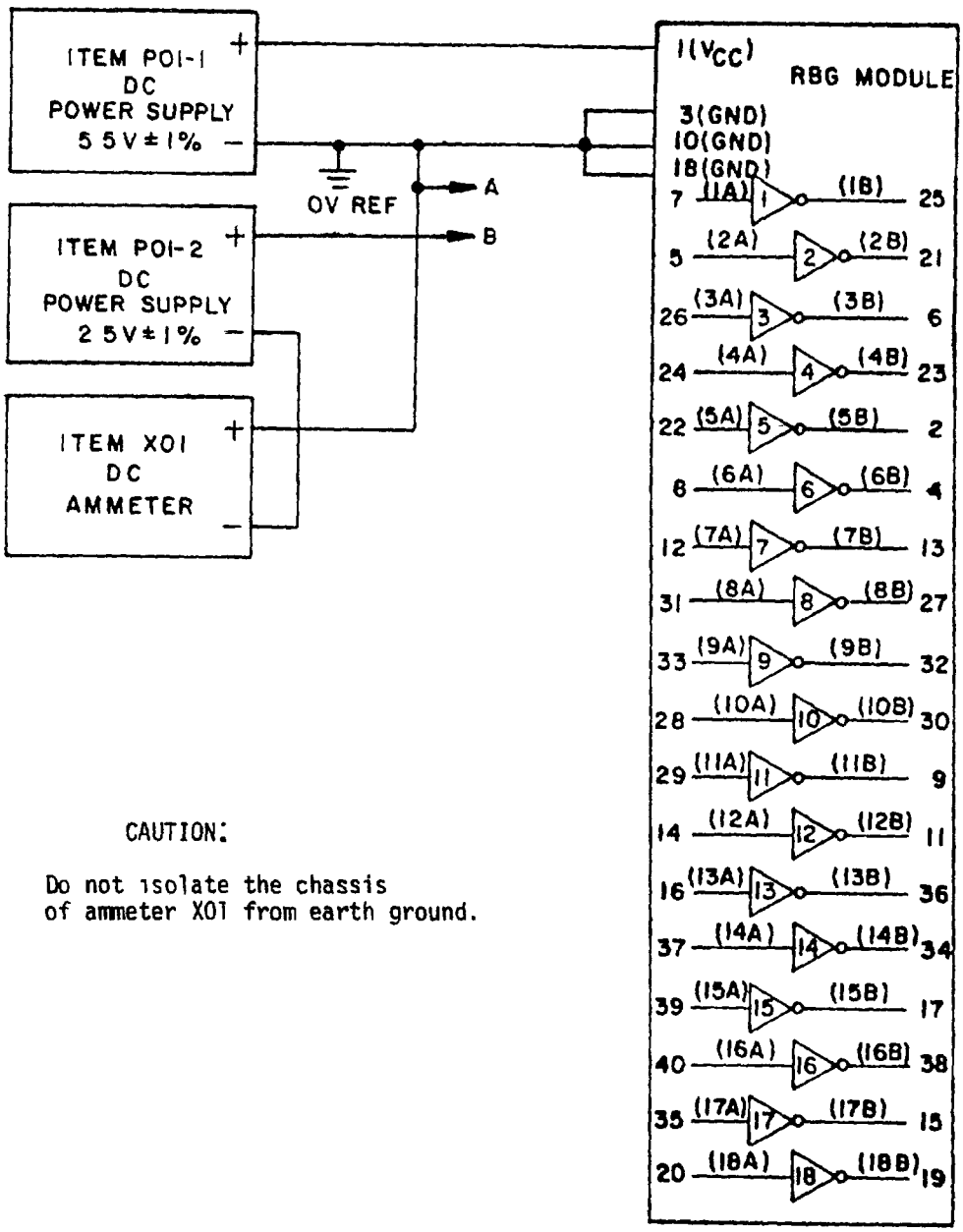
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	1A Input current, high level	5,8,12,14,16,20,22,24, 26,28,29,31,33,35,37,39, 40	7	0	50	µA	13
J2	2A	7,8,12,14,16,20,22,24, 26,28,29,31,33,35,37,39, 40	5				
J3	3A	5,7,8,12,14,16,20,22,24, 28,29,31,33,35,37,39,40	26				
J4	4A	5,7,8,12,14,16,20,22,26, 28,29,31,33,35,37,39,40	24				
J5	5A	5,7,8,12,14,16,20,24,26, 28,29,31,33,35,37,39,40	22				
J6	6A	5,7,12,14,16,20,22,24, 26,28,29,31,33,35,37,39, 40	8				
J7	7A	5,7,8,14,16,20,22,24,26, 28,29,31,33,35,37,39,40	12				
J8	8A	5,7,8,12,14,16,20,22,24, 26,28,29,33,35,37,39,40	31				
J9	9A	5,7,8,12,14,16,20,22,24, 26,28,29,31,35,37,39,40	33				
J10	10A	5,7,8,12,14,16,20,22,24, 26,29,31,33,35,37,39,40	28				
J11	11A Input current, high level	5,7,8,12,14,16,20,22,24, 26,28,31,33,35,37,39,40	29	0	50	µA	13

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J12	12A Input current, high level	5,7,8,12,16,20,22,24,26, 28,29,31,33,35,37,39,40	14	0	50	µA	13
J13	13A	5,7,8,12,14,20,22,24,26, 28,29,31,33,35,37,39,40	16				
J14	14A	5,7,8,12,14,16,20,22,24, 26,28,29,31,33,35,39,40	37				
J15	15A	5,7,8,12,14,16,20,22,24, 26,28,29,31,33,35,37,40	39				
J16	16A	5,7,8,12,14,16,20,22,24, 26,28,29,31,33,35,37,39	40				
J17	17A	5,7,8,12,14,16,20,22,24, 26,28,29,31,33,37,39,40	35				
J18	18A Input current, high level	5,7,8,12,14,16,22,24,26, 28,29,31,33,35,37,39,40	20	0	50	µA	13

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 13. Input current (high-level) test circuit for table XI.

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TABLE XII. Input current (low-level) test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
K1	1A	Input current, low level	7	-0.4	-2.2	mA	14
K2	2A		5				
K3	3A		26				
K4	4A		24				
K5	5A		22				
K6	6A		8				
K7	7A		12				
K8	8A		31				
K9	9A		33				
K10	10A		28				
K11	11A		29				
K12	12A		14				
K13	13A		16				
K14	14A		37				
K15	15A		39				
K16	16A		40				
K17	17A		35				
K18	18A	Input current, low level	20	-0.4	-2.2	mA	14

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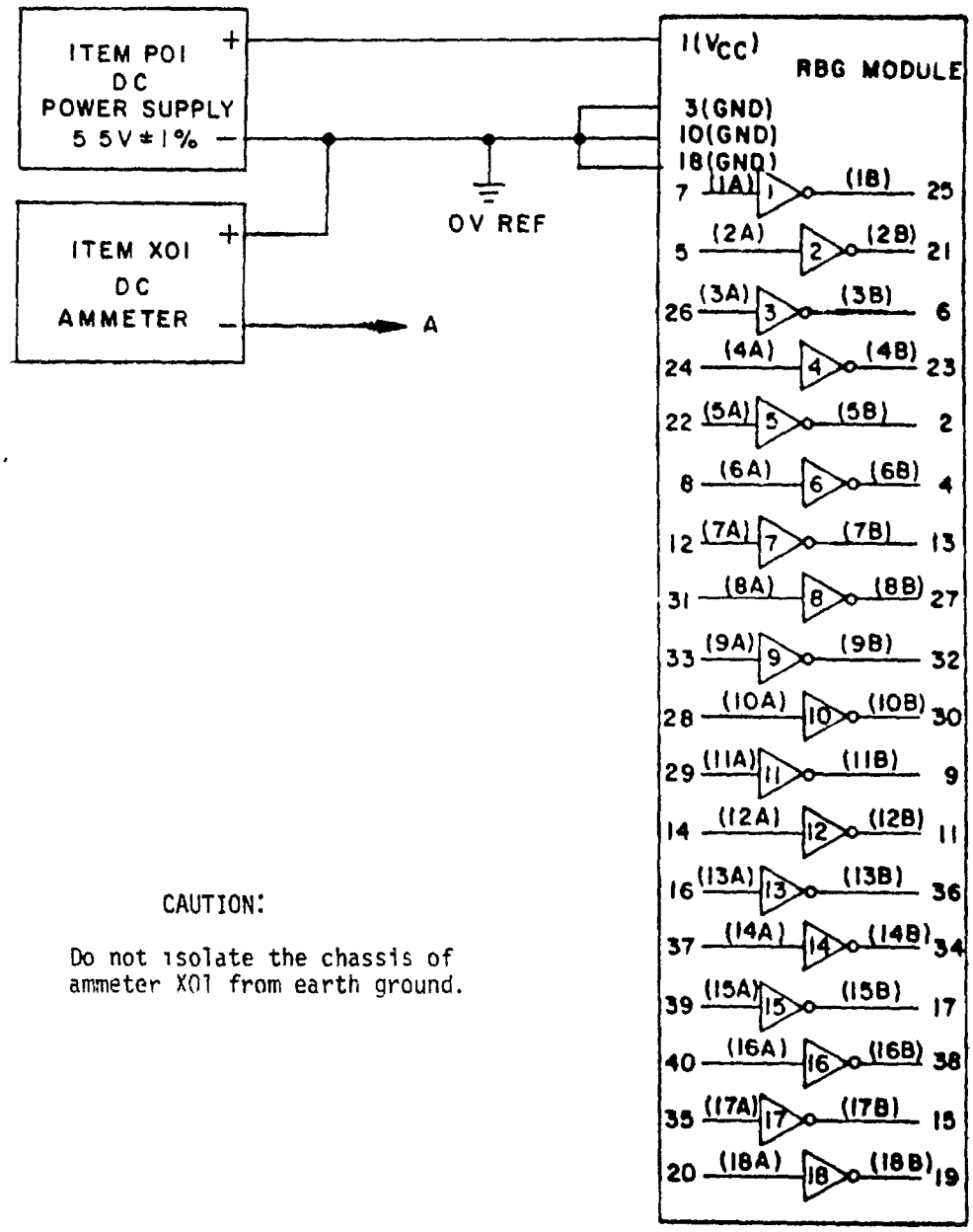


FIGURE 14. Input current (low-level) test circuit for table XII.

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TABLE XIII. Ground pin continuity test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
L1	Ground pin continuity	3	0.0	1.0	Ω	15
L2		18		1.0		
L3		3		0.10		
L4	Ground pin continuity	18	0.0	0.10	Ω	15

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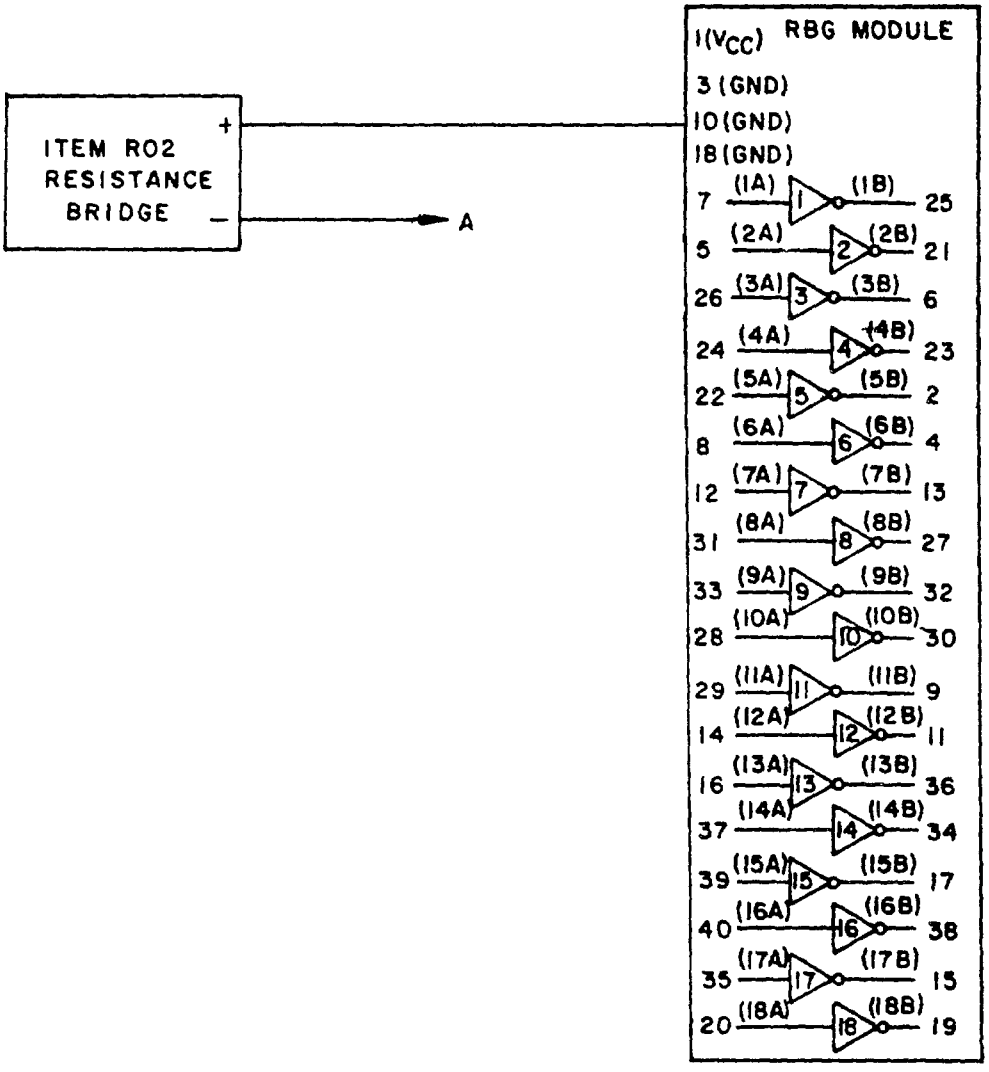


FIGURE 15. Ground pin continuity test circuit for table XIII.

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TABLE XIV. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Power supply current, low level	C	Open	30	162	mA	16
M2	Power supply current, high level	Open	C	15	72	mA	16

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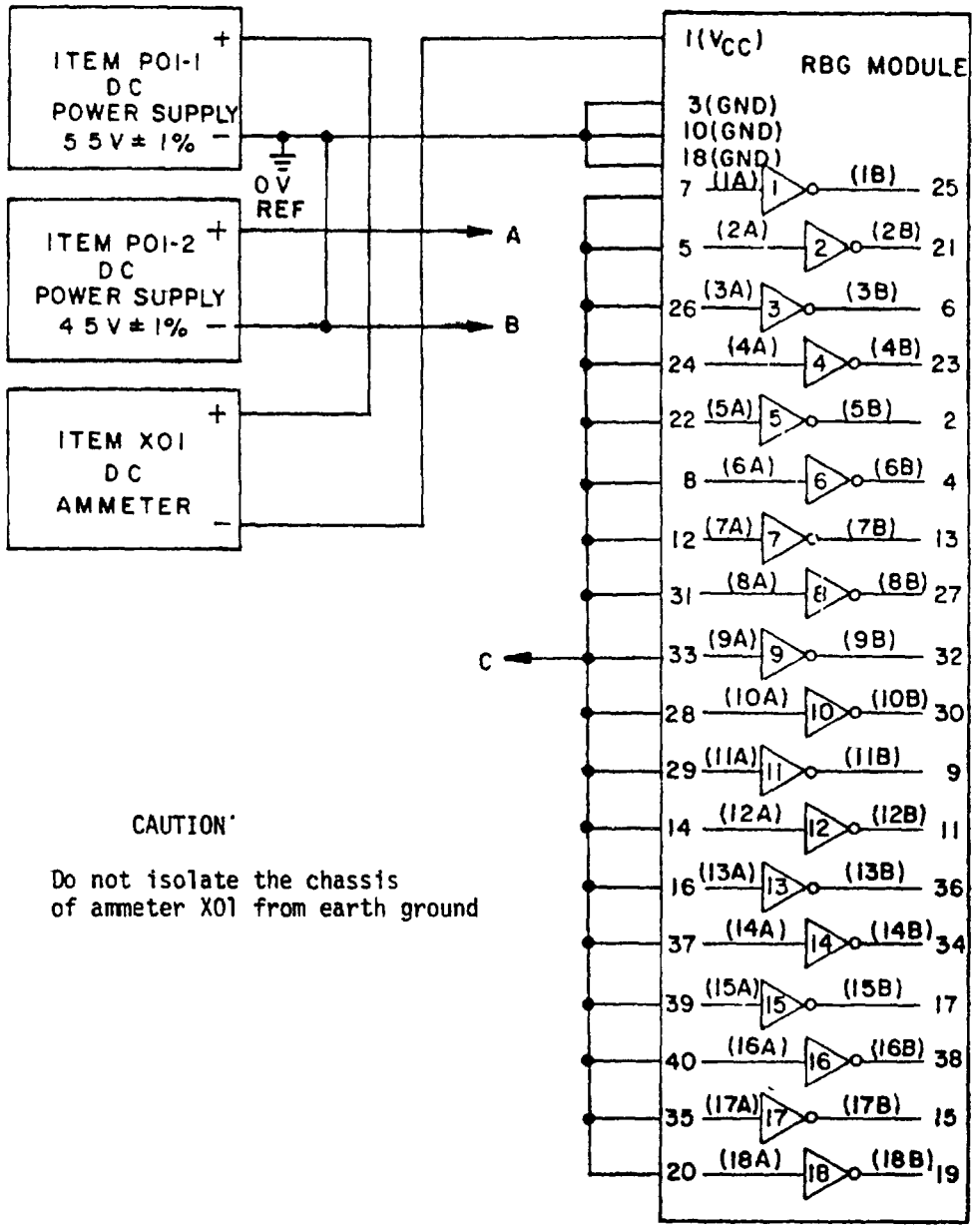


FIGURE 16. Power supply current (low-level and high-level) test circuit for table XIV.

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TABLE XV. Transition time and propagation delay time test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
N1	1B	Transition time (t_{TLH}), C_L = 50 pF	7	25	Open	2.0	15.0	ns	17,18
N2	2B		5	21					
N3	3B		26	6					
N4	4B		24	23					
N5	5B		22	2					
N6	6B		8	4					
N7	7B		12	13					
N8	8B		31	27					
N9	9B		33	32					
N10	10B		28	30					
N11	11B		29	9					
N12	12B		14	11					
N13	13B		16	36					
N14	14B		37	34					
N15	15B		39	17					
N16	16B		40	38					
N17	17B		35	15					
N18	18B	Transition time (t_{TLH}), C_L = 50 pF	20	19	Open	2.0	15.0	ns	17,18

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TABLE XV. Transition time and propagation delay time tests - Cont'd.

Test condition		Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
N19	1B	Transition time (t_{TLH}), C_L = 150 pF	7	25	Open	2.0	27.0	ns	17,18
N20	2B		5	21					
N21	3B		26	6					
N22	4B		24	23					
N23	5B		22	2					
N24	6B		8	4					
N25	7B		12	13					
N26	8B		31	27					
N27	9B		33	32					
N28	10B		28	30					
N29	11B		29	9					
N30	12B		14	11					
N31	13B		16	36					
N32	14B		37	34					
N33	15B		39	17					
N34	16B		40	38					
N35	17B		35	15					
N36	18B	Transition time (t_{TLH}), C_L = 150 pF	20	19	Open	2.0	27.0	ns	17,18

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TABLE XV. Transition time and propagation delay time test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P1	1B	Transition time (t_{THL}), C_L = 50 pF	7	25	5,8,12,14,16,20,22,24, 26,28,29,31,33,35,37, 39,40	2.0	6.0	ns	17,18
P2	2B		5	21	7,8,12,14,16,20,22,24, 26,28,29,31,33,35,37, 39,40				
P3	3B		26	6	5,7,8,12,14,16,20,22, 24,28,29,31,33,35,37, 39,40				
P4	4B		24	23	5,7,8,12,14,16,20,22, 26,28,29,31,33,35,37, 39,40				
P5	5B		22	2	5,7,8,12,14,16,20,24, 26,28,29,31,33,35,37, 39,40				
P6	6B		8	4	5,7,12,14,16,20,22,24, 26,28,29,31,33,35,37, 39,40				
P7	7B		12	13	5,7,8,14,16,20,22,24, 26,28,29,31,33,35,37, 39,40				
P8	8B		31	27	5,7,8,12,14,16,20,22, 24,26,28,29,33,35,37, 39,40				
P9	9B	Transition time (t_{THL}), C_L = 50 pF	33	32	5,7,8,12,14,16,20,22, 24,26,28,29,31,35,37, 39,40	2.0	6.0	ns	17,18

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TABLE XV. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P10	10B Transition time (t_{THL}), C_L = 50 pF	28	30	5,7,8,12,14,16,20,22, 24,26,29,31,33,35,37, 39,40	2.0	6.0	ns	17,18
P11	11B	29	9	5,7,8,12,14,16,20,22, 24,26,28,31,33,35,37, 39,40				
P12	12B	14	11	5,7,8,12,16,20,22,24, 26,28,29,31,33,35,37, 39,40				
P13	13B	16	36	5,7,8,12,14,20,22,24, 26,28,29,31,33,35,37, 39,40				
P14	14B	37	34	5,7,8,12,14,16,20,22, 24,26,28,29,31,33,35, 39,40				
P15	15B	39	17	5,7,8,12,14,16,20,22, 24,26,28,29,31,33,35, 37,40				
P16	16B	40	38	5,7,8,12,14,16,20,22, 24,26,28,29,31,33,35, 37,39				
P17	17B	35	15	5,7,8,12,14,16,20,22, 24,26,28,29,31,33,37, 39,40				
P18	18B Transition time (t_{THL}), C_L = 50 pF	20	19	5,7,8,12,14,16,22,24, 26,28,29,31,33,35,37, 39,40	2.0	6.0	ns	17,18

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TABLE XV. Transition time and propagation delay time test - Continued.

Test condition		Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P19	1B	Transition time (t_{THL}), C_L = 150 pF	7	25	5,8,12,14,16,20,22,24, 26,28,29,31,33,35,37, 39,40	2.0	11.0	ns	17,18
P20	2B		5	21	7,8,12,14,16,20,22,24, 26,28,29,31,33,35,37, 39,40				
P21	3B		26	6	5,7,8,12,14,16,20,22, 24,28,29,31,33,35,37, 39,40				
P22	4B		24	23	5,7,8,12,14,16,20,22, 26,28,29,31,33,35,37, 39,40				
P23	5B		22	2	5,7,8,12,14,16,20,24, 26,28,29,31,33,35,37, 39,40				
P24	6B		8	4	5,7,12,14,16,20,22,24, 26,28,29,31,33,35,37, 39,40				
P25	7B		12	13	5,7,8,14,16,20,22,24, 26,28,29,31,33,35,37, 39,40				
P26	8B		31	27	5,7,8,12,14,16,20,22, 24,26,28,29,33,35,37, 39,40				
P27	9B	Transition time (t_{THL}), C_L = 150 pF	33	32	5,7,8,12,14,16,20,22, 24,26,28,29,31,35,37, 39,40	2.0	11.0	ns	17,18

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TABLE XV. Transition time and propagation delay time test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P28	10B	Transition time (t_{THL}), C_L = 150 pF	28	30	5,7,8,12,14,16,20,22, 24,26,29,31,33,35,37, 39,40	2.0	11.0	ns	17,18
P29	11B		29	9	5,7,8,12,14,16,20,22, 24,26,28,31,33,35,37, 39,40				
P30	12B		14	11	5,7,8,12,16,20,22,24, 26,28,29,31,33,35,37, 39,40				
P31	13B		16	36	5,7,8,12,14,20,22,24, 26,28,29,31,33,35,37, 39,40				
P32	14B		37	34	5,7,8,12,14,16,20,22, 24,26,28,29,31,33,35, 39,40				
P33	15B		39	17	5,7,8,12,14,16,20,22, 24,26,28,29,31,33,35, 37,40				
P34	16B		40	38	5,7,8,12,14,16,20,22, 24,26,28,29,31,33,35, 37,39				
P35	17B		35	15	5,7,8,12,14,16,20,22, 24,26,28,29,31,33,37, 39,40				
P36	18B	Transition time (t_{THL}), C_L = 150 pF	20	19	5,7,8,12,14,16,22,24, 26,28,29,31,33,35,37, 39,40	2.0	11.0	ns	17,18

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TABLE XV. Transition time and propagation delay time test - Continued.

	Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
*	Q1	1B Propagation delay time (t_{PLH}), C_L = 50 pF	7	25	Open	2.0	7.0	ns	17,18
	Q2	2B	5	21					
	Q3	3B	26	6					
	Q4	4B	24	23					
	Q5	5B	22	2					
	Q6	6B	8	4					
	Q7	7B	12	13					
	Q8	8B	31	27					
	Q9	9B	33	32					
	Q10	10B	28	30					
	Q11	11B	29	9					
	Q12	12B	14	11					
	Q13	13B	16	36					
	Q14	14B	37	34					
	Q15	15B	39	17					
	Q16	16B	40	38					
	Q17	17B	35	15					
*	Q18	18B Propagation delay time (t_{PLH}), C_L = 50 pF	20	19	Open	2.0	7.0	ns	17,18

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TABLE XV. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q19	1B Propagation delay time (t_{PLH}), C_L = 150 pF	7	25	Open	4.0	10.5	ns	17,18
Q20	2B	5	21					
Q21	3B	26	6					
Q22	4B	24	23					
Q23	5B	22	2					
Q24	6B	8	4					
Q25	7B	12	13					
Q26	8B	31	27					
Q27	9B	33	32					
Q28	10B	28	30					
Q29	11B	29	9					
Q30	12B	14	11					
Q31	13B	16	36					
Q32	14B	37	34					
Q33	15B	39	17					
Q34	16B	40	38					
Q35	17B	35	15					
Q36	18B Propagation delay time (t_{PLH}), C_L = 150 pF	20	19	Open	4.0	10.5	ns	17,18

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TABLE XV. Transition time and propagation delay time test - Continued.

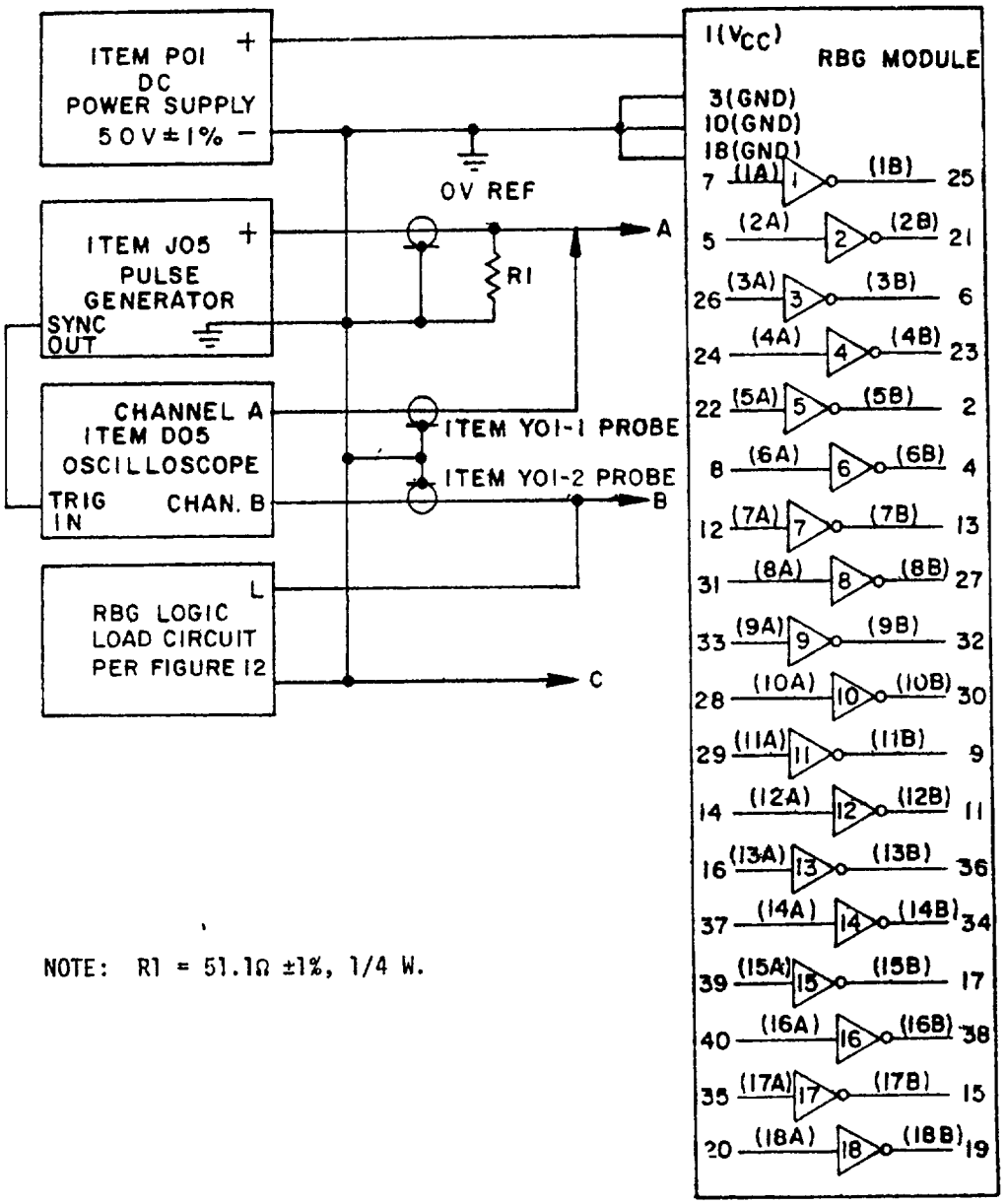
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	1B Propagation delay time (t_{PHL}), C_L = 50 pF	7	25	Open	2.0	7.5	ns	17,18
R2	2B	5	21					
R3	3B	26	6					
R4	4B	24	23					
R5	5B	22	2					
R6	6B	8	4					
R7	7B	12	13					
R8	8B	31	27					
R9	9B	33	32					
R10	10B	28	30					
R11	11B	29	9					
R12	12B	14	11					
R13	13B	16	36					
R14	14B	37	34					
R15	15B	39	17					
R16	16B	40	38					
R17	17B	35	15					
R18	18B Propagation delay time (t_{PHL}), C_L = 50 pF	20	19	Open	2.0	7.5	ns	17,18

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TABLE XV. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R19	1B Propagation delay time (t_{PHL}), C_L = 150 pF	7	25	Open	4.0	11.5	ns	17,18
R20	2B	5	21					
R21	3B	26	6					
R22	4B	24	23					
R23	5B	22	2					
R24	6B	8	4					
R25	7B	12	13					
R26	8B	31	27					
R27	9B	33	32					
R28	10B	28	30					
R29	11B	29	9					
R30	12B	14	11					
R31	13B	16	36					
R32	14B	37	34					
R33	15B	39	17					
R34	16B	40	38					
R35	17B	35	15					
R36	18B Propagation delay time (t_{PHL}), C_L = 150 pF	20	19	Open	4.0	11.5	ns	17,18

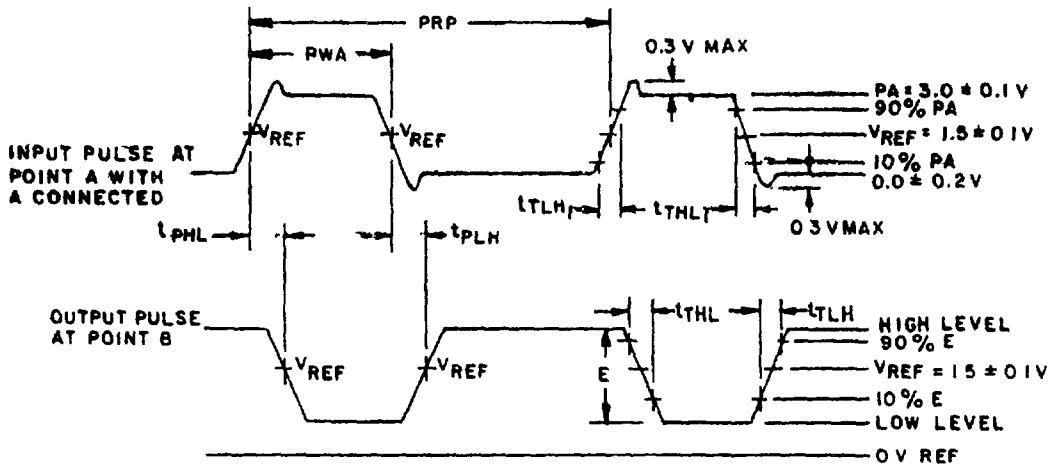
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NOTE: R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 17. Transition time and propagation delay time test circuit for table XV.

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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 4 \pm 1$ ns.
   PWA = 100 ±10 ns.
   PRP = 500 ±50 ns.
2. Before making the measurements specified in table XV, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pin; between probes YO1-1 -and YO1-2 and input and output pins; and between load circuit and output pin, shall be less than 1 inch (25.4 mm)
4. All voltages are referenced to ground.

FIGURE 18. Transition time and propagation delay time test circuit waveforms for table XV.

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TABLE XVI. Output to output delta propagation delay time test.

Test condition	Parameter		Connect			Initial limits 25° +5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
S1	1B to 2B	Output to output delta propagation delay time (Δt_{PLH}), C_L = 50 pF	5,7,26	25	21	-3.0	3.0	ns	19,20
S2	2B to 3B		5,7,26	21	6				
S3	1B to 3B		5,7,26	25	6				
S4	4B to 5B		8,22,24	23	2				
S5	5B to 6B		8,22,24	2	4				
S6	4B to 6B		8,22,24	23	4				
S7	7B to 8B		12,31,33	13	27				
S8	8B to 9B		12,31,33	27	32				
S9	7B to 9B		12,31,33	13	32				
S10	10B to 11B		14,28,29	30	9				
S11	11B to 12B		14,28,29	9	11				
S12	10B to 12B		14,28,29	30	11				
S13	13B to 14B		16,37,39	36	34				
S14	14B to 15B		16,37,39	34	17				
S15	13B to 15B		16,37,39	36	17				
S16	16B to 17B	Output to output delta propagation delay time (Δt_{PLH}), C_L = 50 pF	20,35,40	38	15	-3.0	3.0	ns	19,20

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TABLE XVI. Output to output delta propagation delay time test - Continued.

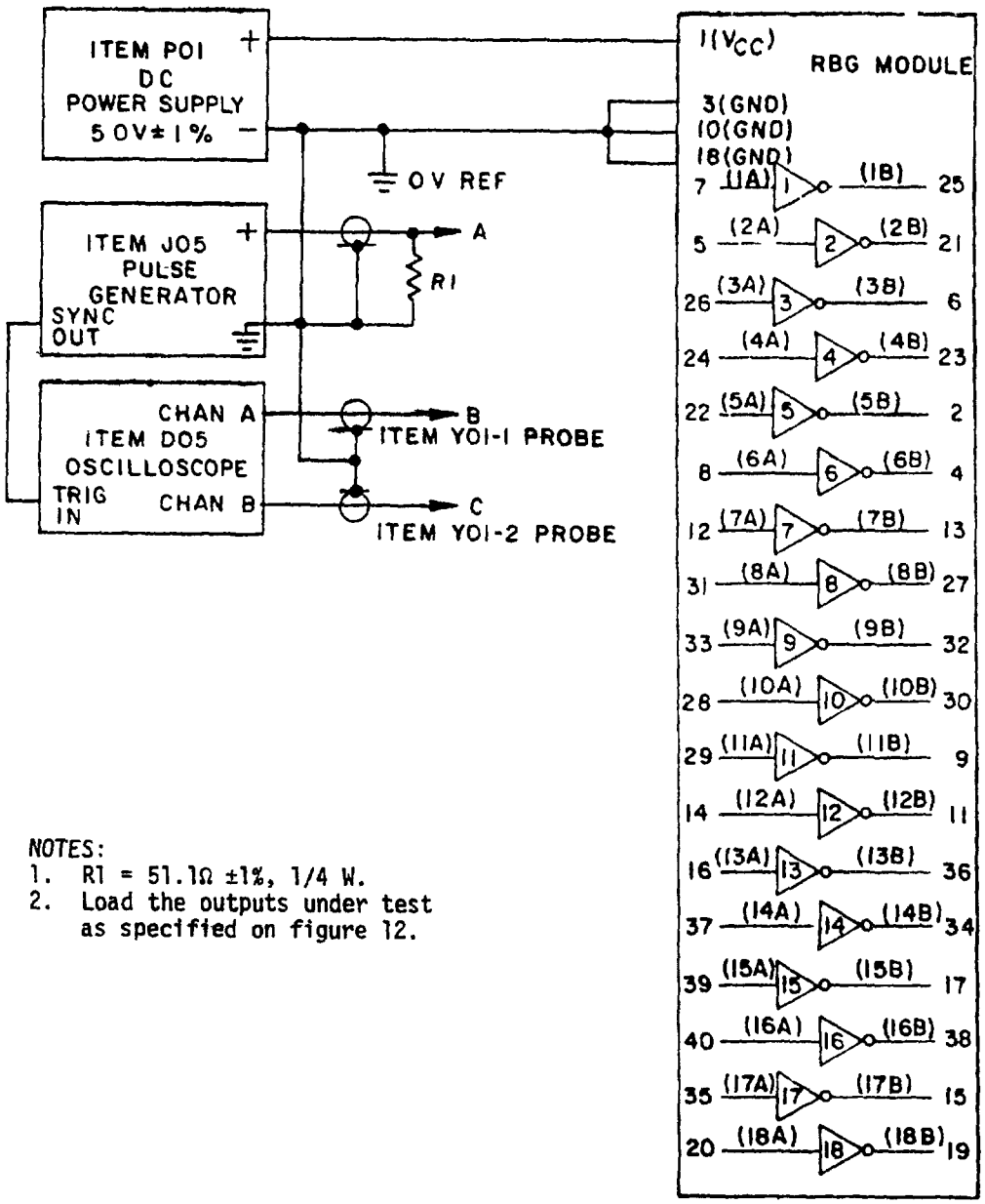
Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
S17	17B to 18B	Output to output delta propagation delay time (Δt_{PLH}), C_L = 50 pF	20,35,40	15	19	-3.0	3.0	ns	19,20
S18	16B to 18B	Output to output delta propagation delay time (Δt_{PLH}), C_L = 50 pF	20,35,40	38	19				
T1	1B to 2B	Output to output delta propagation delay time (Δt_{PHL}), C_L = 50 pF	5,7,26	25	21				
T2	2B to 3B		5,7,26	21	6				
T3	1B to 3B		5,7,26	25	6				
T4	4B to 5B		8,22,24	23	2				
T5	5B to 6B		8,22,24	2	4				
T6	4B to 6B		8,22,24	23	4				
T7	7B to 8B		12,31,33	13	27				
T8	8B to 9B		12,31,33	27	32				
T9	7B to 9B		12,31,33	13	32				
T10	10B to 11B	Output to output delta propagation delay time (Δt_{PHL}), C_L = 50 pF	14,28,29	30	9	-3.0	3.0	ns	19,20

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TABLE XVI. Output to output delta propagation delay time test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T11	11B to 12B	Output to output delta propagation delay time (Δt_{PHL}), C_L = 50 pF	14,28,29	9	11	-3.0	3.0	ns	19,20
T12	10B to 12B		14,28,29	30	11				
T13	13B to 14B		16,37,39	36	34				
T14	14B to 15B		16,37,39	34	17				
T15	13B to 15B		16,37,39	36	17				
T16	16B to 17B		20,35,40	38	15				
T17	17B to 18B		20,35,40	15	19				
T18	16B to 18B	Output to output delta propagation delay time (Δt_{PHL}), C_L = 50 pF	20,35,40	38	19	-3.0	3.0	ns	19,20

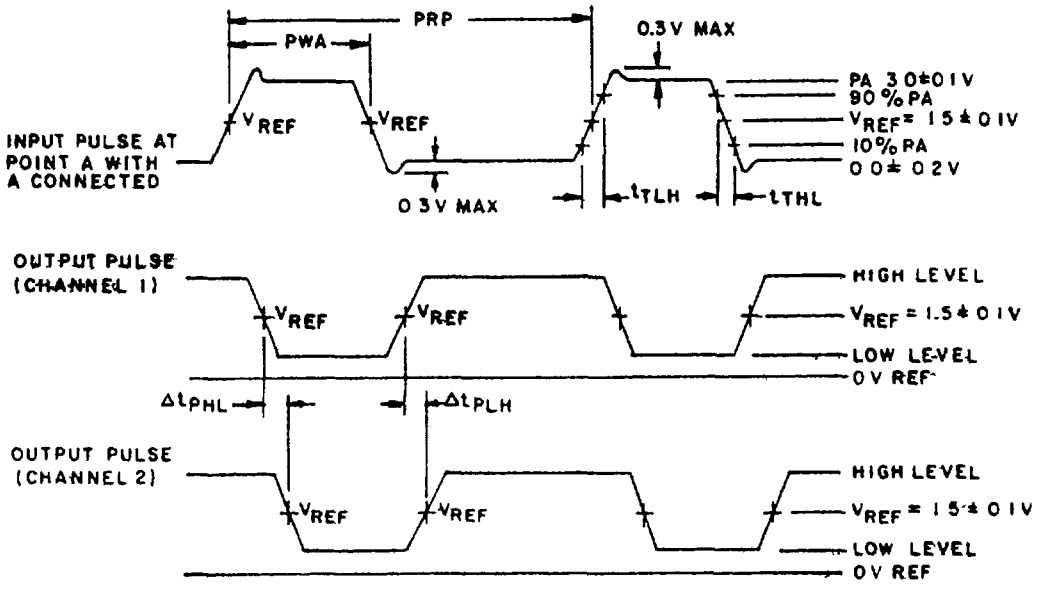
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load the outputs under test as specified on figure 12.

FIGURE 19. Output to output delta propagation delay time test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   PWA = 100 ±10 ns.
   PRP = 500 ±50 ns.
2. Before making the measurements specified in table XVI, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pins; between probes YO1-1 and YO1-2 and output pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 20. Output to output delta propagation delay time test circuit waveforms for table XVI.

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TABLE XVII. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	1B Output crosstalk, low level	7	Open	5,8,12, 14,16, 20,22, 24,26, 28,29, 31,33, 35,37, 39,40	25	0	800	mV	21,22
U2	2B	5		7,8,12, 14,16, 20,22, 24,26, 28,29, 31,33, 35,37, 39,40	21				
U3	3B	26		5,7,8, 12,14, 16,20, 22,24, 28,29, 31,33, 35,37, 39,40	6				
U4	4B Output crosstalk, low level	24	Open	5,7,8, 12,14, 16,20, 22,26, 28,29, 31,33, 35,37, 39,40	23	0	800	mV	21,22

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U5	5B Output crosstalk, low level	22	Open	5,7,8, 12,14, 16,20, 24,26, 28,29, 31,33, 35,37, 39,40	2	0	800	mV	21,22
U6	6B	8		5,7,12, 14,16, 20,22, 24,26, 28,29, 31,33, 35,37, 39,40	4				
U7	7B	12		5,7,8, 14,16, 20,22, 24,26, 28,29, 31,33, 35,37, 39,40	13				
U8	8B Output crosstalk, low level	31	Open	5,7,8, 12,14, 16,20, 22,24, 26,28, 29,33, 35,37, 39,40	27	0	800	mV	21,22

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued

Test condition	Parameter	Connect				Initial limits 25° +5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U9	9B Output crosstalk, low level	33	Open	5,7,8, 12,14, 16,20, 22,24, 26,28, 29,31, 35,37, 39,40	32	0	800	mV	21,22
U10	10B	28		5,7,8, 12,14, 16,20, 22,24, 26,29, 31,33, 35,37, 39,40	30				
U11	11B	29		5,7,8, 12,14, 16,20, 22,24, 26,28, 31,33, 35,37, 39,40	9				
U12	12B Output crosstalk, low level	14	Open	5,7,8, 12,16, 20,22, 24,26, 28,29, 31,33, 35,37, 39,40	11	0	800	mV	21,22

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U13	13B Output crosstalk, low level	16	Open	5,7,8, 12,14, 20,22, 24,26, 28,29, 31,33, 35,37, 39,40	36	0	800	mV	21,22
U14	14B	37		5,7,8, 12,14, 16,20, 22,24, 26,28, 29,31, 33,35, 39,40	34				
U15	15B	39		5,7,8, 12,14, 16,20, 22,24, 26,28, 29,31, 33,35, 37,40	17				
U16	16B Output crosstalk, low level	40	Open	5,7,8, 12,14, 16,20, 22,24, 26,28, 29,31, 33,35, 37,39	38	0	800	mV	21,22

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U17	17B Output crosstalk, low level	35	Open	5,7,8, 12,14, 16,20, 22,24, 26,28, 29,31, 33,37, 39,40	15	0	800	mV	21,22
U18	18B	20		5,7,8, 12,14, 16,22, 24,26, 28,29, 31,33, 35,37, 39,40	19		800		
U19	1B	5,7,26		8,12, 14,16, 20,22, 24,28, 29,31, 33,35, 37,39, 40	25		525		
U20	2B Output crosstalk, low level	5,7,26	Open	8,12, 14,16, 20,22, 24,28, 29,31, 33,35, 37,39, 40	21	0	525	mV	21,22

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U21	3B Output crosstalk, low level	5,7, 26	Open	8,12, 14,16, 20,22, 24,28, 29,31, 33,35, 37,39, 40	6	0	525	mV	21,22
U22	4B	8,22, 24		5,7,12, 14,16, 20,26, 28,29, 31,33, 35,37, 39,40	23				
U23	5B	8,22, 24		5,7,12, 14,16, 20,26, 28,29, 31,33, 35,37, 39,40	2				
U24	6B Output crosstalk, low level	8,22, 24	Open	5,7,12, 14,16, 20,26, 28,29, 31,33, 35,37, 39,40	4	0	525	mV	21,22

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U25	7B Output crosstalk, low level	12,31, 33	Open	5,7,8, 14,16, 20,22, 24,26, 28,29, 35,37, 39,40	13	0	525	mV	21,22
U26	8B	12,31, 33		5,7,8, 14,16, 20,22, 24,26, 28,29, 35,37, 39,40	27				
U27	9B	12,31, 33		5,7,8, 14,16, 20,22, 24,26, 28,29, 35,37, 39,40	32				
U28	10B Output crosstalk, low level	14,28, 29	Open	5,7,8, 12,16, 20,22, 24,26, 31,33, 35,37, 39,40	30	0	525	mV	21,22

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U29	11B Output crosstalk, low level	14,28, 29	Open	5,7,8, 12,16, 20,22, 24,26, 31,33, 35,37, 39,40	9	0	525	mV	21,22
U30	12B	14,28, 29		5,7,8, 12,16, 20,22, 24,26, 31,33, 35,37, 39,40	11				
U31	13B	16,37, 39		5,7,8, 12,14, 20,22, 24,26, 28,29, 31,33, 35,40	36				
U32	14B Output crosstalk, low level	16,37, 39	Open	5,7,8, 12,14, 20,22, 24,26, 28,29, 31,33, 35,40	34	0	525	mV	21,22

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U33	15B Output crosstalk, low level	16,37, 39	Open	5,7,8, 12,14, 20,22, 24,26, 28,29, 31,33, 35,40	17	0	525	mV	21,22
U34	16B	20,35, 40		5,7,8, 12,14, 16,22, 24,26, 28,29, 31,33, 37,39	38				
U35	17B	20,35, 40		5,7,8, 12,14, 16,22, 24,26, 28,29, 31,33, 37,39	15				
U36	18B Output crosstalk, low level	20,35, 40	Open	5,7,8 12,14, 16,22, 24,26, 28,29, 31,33, 37,39	19	0	525	mV	21,22

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	1B Output crosstalk, high level	Open	7	5,8,12, 14,16, 20,22, 24,26, 28,29, 31,33, 35,37, 39,40	25	0	400	mV	21,22
V2	2B		5	7,8,12, 14,16, 20,22, 24,26, 28,29, 31,33, 35,37, 39,40	21				
V3	3B		26	5,7,8, 12,14, 16,20, 22,24, 28,29, 31,33, 35,37, 39,40	6				
V4	4B Output crosstalk, high level	Open	24	5,7,8, 12,14, 16,20, 22,26, 28,29, 31,33, 35,37, 39,40	23	0	400	mV	21,22

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V5	5B Output crosstalk, high level	Open	22	5,7,8, 12,14, 16,20, 24,26, 28,29, 31,33, 35,37, 39,40	2	0	400	mV	21,22
V6	6B		8	5,7,12, 14,16, 20,22, 24,26, 28,29, 31,33, 35,37, 39,40	4				
V7	7B		12	5,7,8, 14,16, 20,22, 24,26, 28,29, 31,33, 35,37, 39,40	13				
V8	8B Output crosstalk, high level	Open	31	5,7,8, 12,14, 16,20, 22,24, 26,28, 29,33, 35,37, 39,40	27	0	400	mV	21,22

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V9	9B Output crosstalk, high level	Open	33	5,7,8, 12,14, 16,20, 22,24, 26,28, 29,31, 35,37, 39,40	32	0	400	mV	21,22
V10	10B		28	5,7,8, 12,14, 16,20, 22,24, 26,29, 31,33, 35,37, 39,40	30				
V11	11B		29	5,7,8, 12,14, 16,20, 22,24, 26,28, 31,33, 35,37, 39,40	9				
V12	12B Output crosstalk, high level	Open	14	5,7,8, 12,16, 20,22, 24,26, 28,29, 31,33, 35,37, 39,40	11	0	400	mV	21,22

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

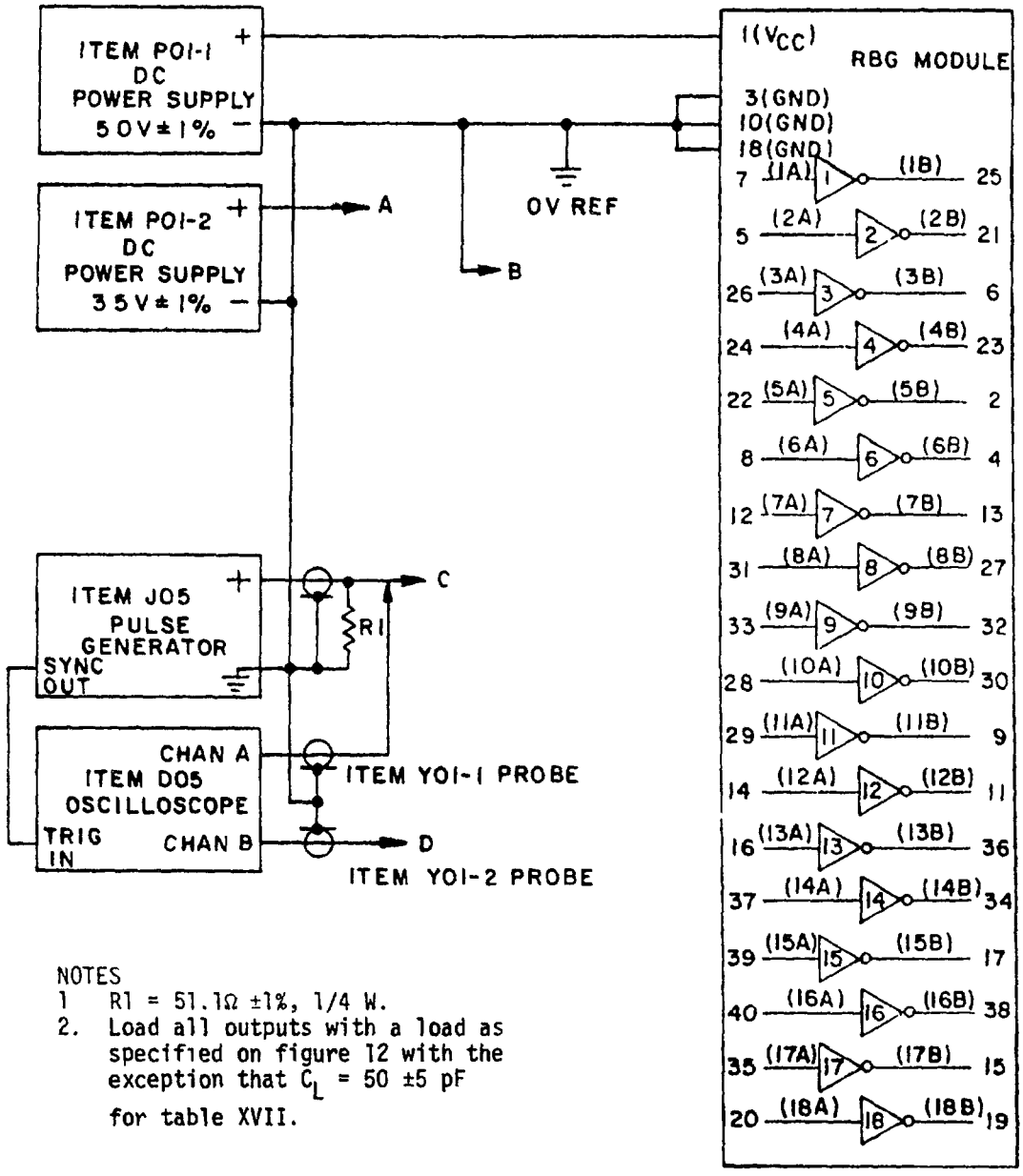
Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V13	13B Output crosstalk, high level	Open	16	5,7,8, 12,14, 20,22, 24,26, 28,29, 31,33, 35,37, 39,40	36	0	400	mV	21,22
V14	14B		37	5,7,8, 12,14, 16,20, 22,24, 26,28, 29,31, 33,35, 39,40	34				
V15	15B		39	5,7,8, 12,14, 16,20, 22,24, 26,28, 29,31, 33,35, 37,40	17				
V16	16B Output crosstalk, high level	Open	40	5,7,8, 12,14, 16,20, 22,24, 26,28, 29,31, 33,35, 37,39	38	0	400	mV	21,22

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V17	17B Output crosstalk, high level	Open	35	5,7,8, 12,14, 16,20, 22,24, 26,28, 29,31, 33,37, 39,40	15	0	400	mV	21,22
V18	18B Output crosstalk, high level	Open	20	5,7,8, 12,14, 16,22, 24,26, 28,29, 31,33, 35,37, 39,40	19	0	400	mV	21,22

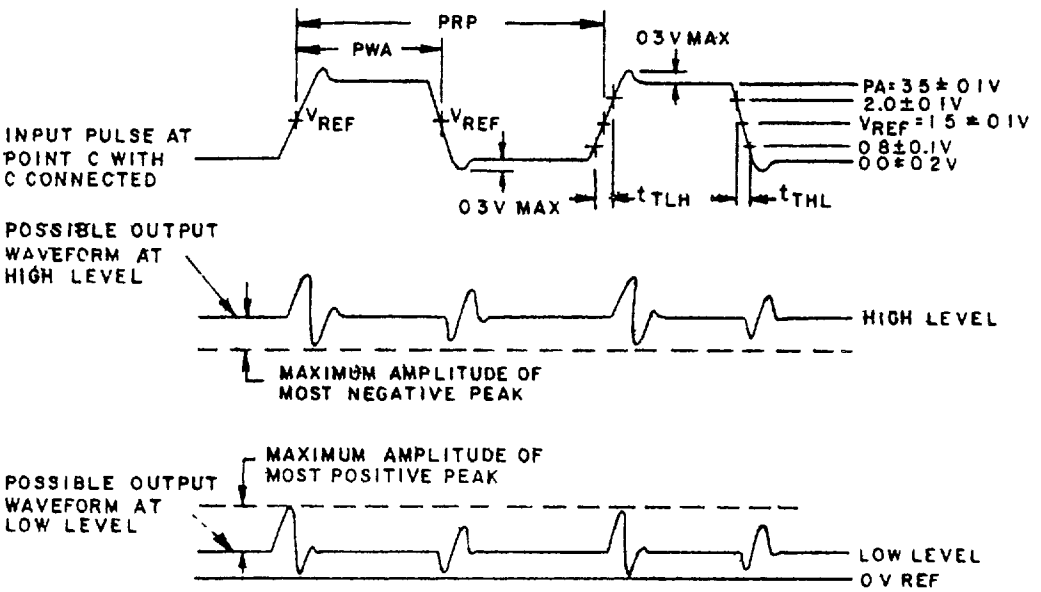
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NOTES
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with a load as specified on figure 12 with the exception that C_L = 50 ±5 pF for table XVII.

FIGURE 21. Output crosstalk (low-level and high-level)test circuit for table XVII.

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NOTES:

1. Input pulse parameters. For each test condition with the module connected to the pulse generator, adjust the pulse generator for the required pulse amplitude, pulse low level voltage, and transition times. Verify the input pulse parameters at each input pin before making each measurement.

   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (measured at the 0.8 V and 2.0 V points of the input pulse)
   PWA = 250 ±50 ns
   PRP = 500 ±100 ns.

2. The lead length of electrical conductors between resistor R1 and input pins; between probes Y01-1 and Y01-2 and input and output pins; between load circuits and output pins; and between ground leads of probes Y01-1 and Y01-2 and test fixture and module ground pins, shall be less than 1.00 inch (25.4 mm)

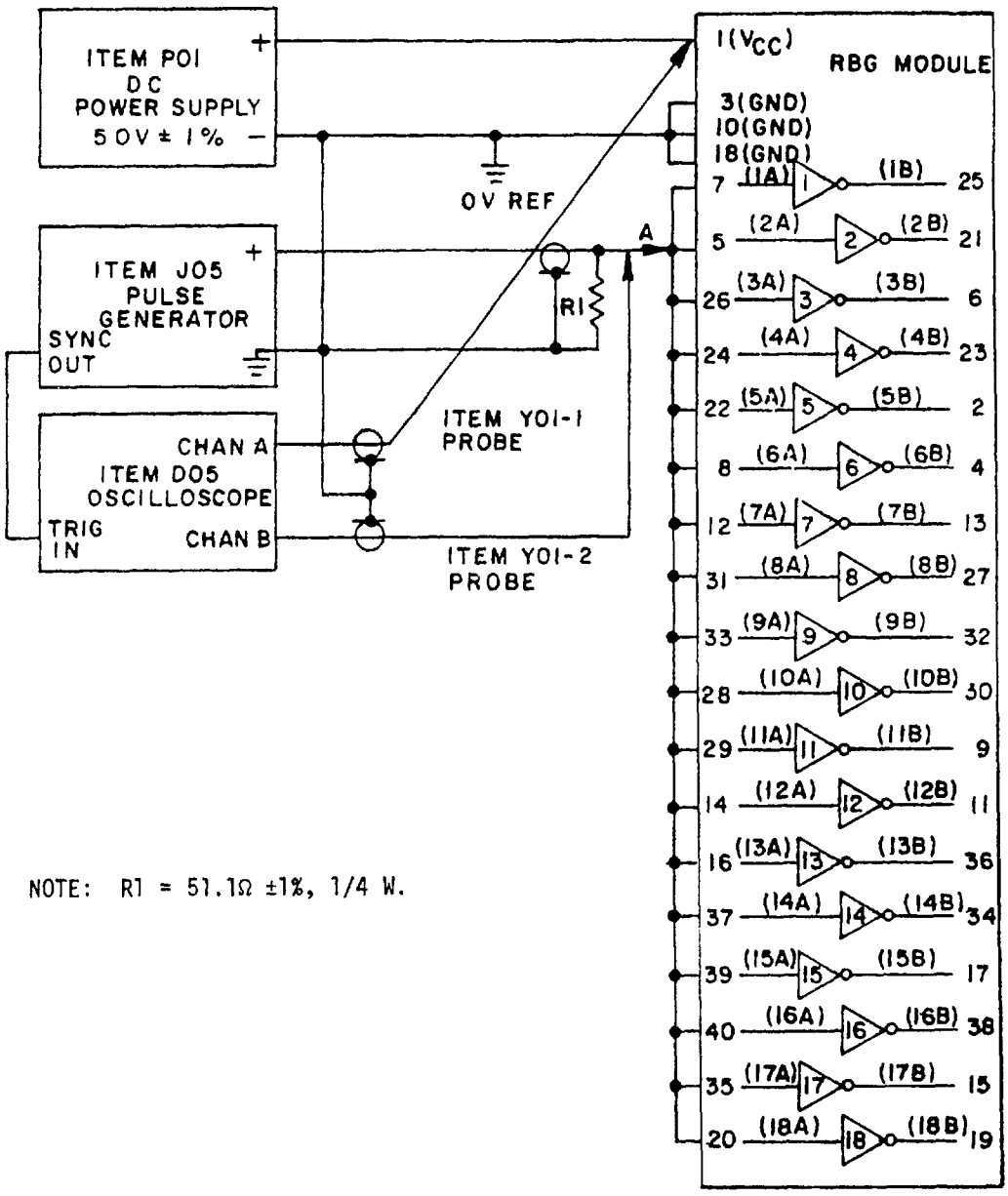
FIGURE 2? Output crosstalk (low-level and high-level) test circuit waveforms for table XVII

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
W	Power supply transient amplitude	0	1.0	V	23,24

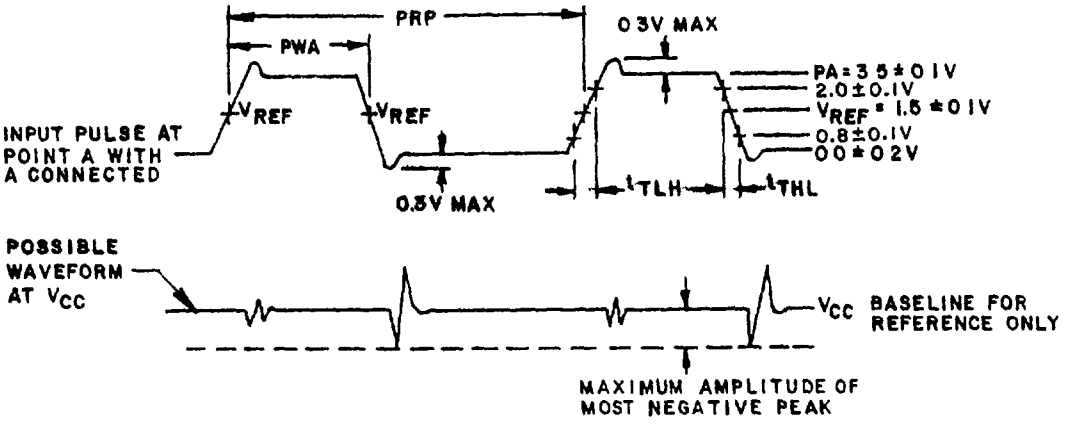
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NOTE: R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 23. Power supply transient amplitude test circuit for table XVIII.

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NOTE: Input pulse parameters: With the module connected to the pulse generator, adjust the pulse generator for the required pulse amplitude, pulse low level voltage and transition times. Verify the input pulse parmeters at each input pin before making the measurement.

t_{TLH} = 7.5 ±2.5 ns, t_{THL} = 5 ±2 ns.
PWA = 500 ±50 ns.
PRP = 1,000 ±200 ns.

FIGURE 24. Power supply transient amplitude test circuit waveforms for table XVIII.

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TABLE XIX. Power supply filter capacitance test.

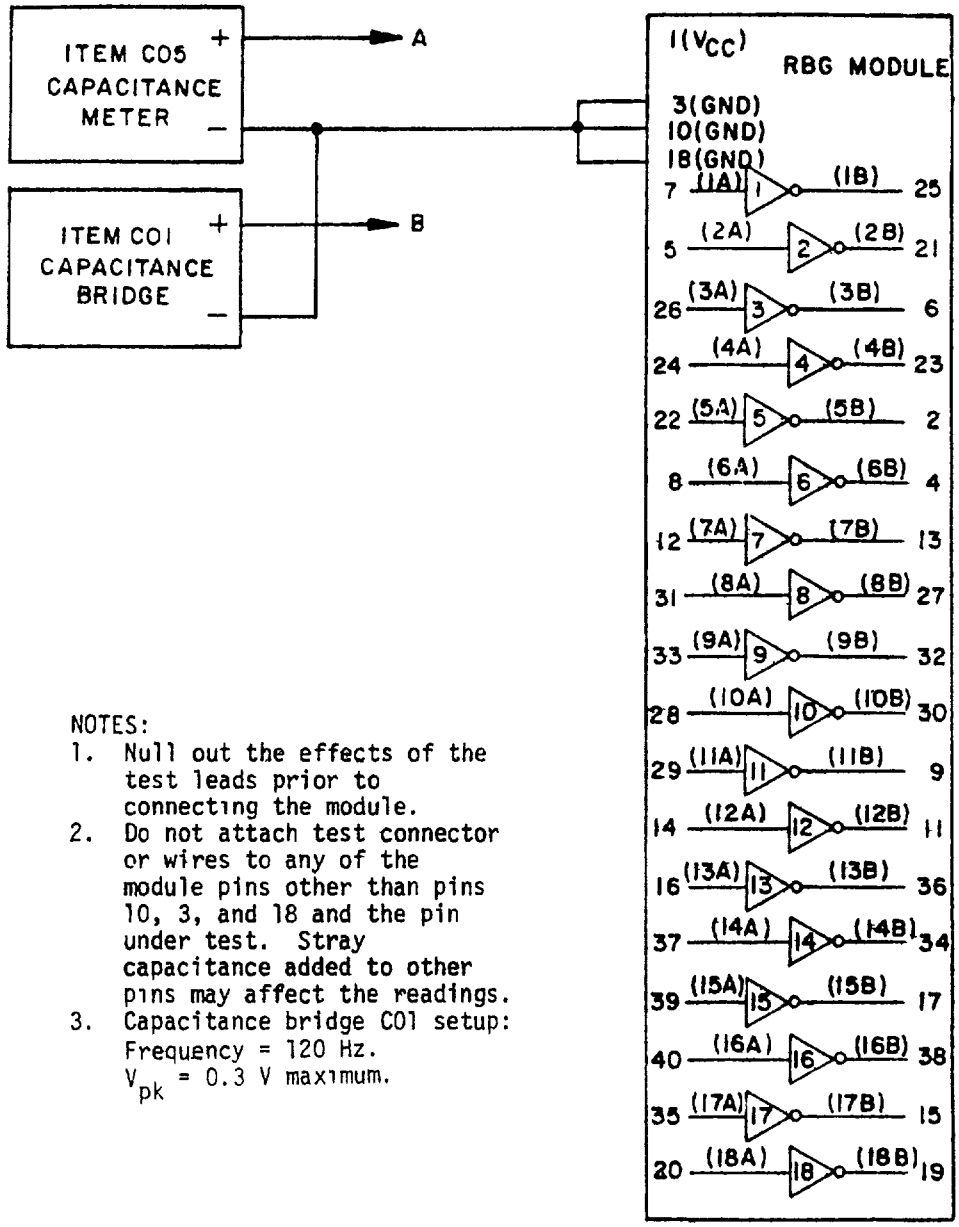
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X	Power supply filter capacitance	Open	1	1.6	3.0	µF	25

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TABLE XX. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y1	1A Input capacitance	7	Open	5.0	15.0	pF	25
Y2	2A	5					
Y3	3A	26					
Y4	4A	24					
Y5	5A	22					
Y6	6A	8					
Y7	7A	12					
Y8	8A	31					
Y9	9A	33					
Y10	10A	28					
Y11	11A	29					
Y12	12A	14					
Y13	13A	16					
Y14	14A	37					
Y15	15A	39					
Y16	16A	40					
Y17	17A	35					
Y18	18A Input capacitance	20	Open	5.0	15.0	pF	25

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pins 10, 3, and 18 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge C01 setup: Frequency = 120 Hz. V_{pk} = 0.3 V maximum.

FIGURE 25. Power supply filter and input capacitance test circuit for tables XIX and XX.

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